

Degree project in Innovation Management & Product Development

Second cycle, 30 credits

An AI-Based Decision-Support Framework for Corporate Boards

In Foresight and Strategic Environmental Scanning

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An AI-Based Decision-Support Framework for Corporate Boards

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6/21/26

Master's Thesis

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Abstract

Corporate boards face four structural information challenges that constrain their capacity for strategic foresight: a management filter that controls what reaches the board, a temporal mismatch between episodic governance and the pace of strategic disruption, a competitive intelligence gap that limits scanning beyond the firm's immediate sector, and a structural optimism in scenario planning. Artificial intelligence offers a possible response to these challenges, yet no design framework exists that specifies how AI-based foresight tools should be designed and governed for the board context specifically. This thesis aims to develop such a framework.

Grounded in Design Science Research methodology, the study combines a seven-stream theoretical review (bounded rationality, corporate foresight, AI in organizational decision-making, technology adoption, trust and explainability, board processes and group information processing, and AI governance and fiduciary accountability) with qualitative empirical data from ten semi-structured interviews with senior board professionals across Nordic governance contexts, supplemented by one expert advisory session on foresight methodology. The thematic analysis produced eight themes across three layers (context, capability, and adoption). Based on these, a design framework of twenty-one design principles across four dimensions is developed: cognitive augmentation (DP1 to DP6), foresight function (DP7 to DP11), trust and adoption (DP12 to DP16), and governance and accountability (DP17 to DP21).

The results indicates that AI has the potential to address all four structural challenges, but only if designed to preserve the deliberative conditions that make boards valuable, deployed through a structured change management protocol, and governed in ways that maintain the fiduciary accountability of directors.

The framework contributes four theoretical advances: it bridges the corporate foresight and AI-in-boards literatures; it extends Forbes and Milliken's (1999) process model of board decision-making into the AI context; it extends Raisch and Krakowski's (2021) automation-augmentation paradox to governance-specific accountability constraints; and it introduces epistemic independence as a governance principle, distinct from and more demanding than existing explainability frameworks.

Keywords

Artificial Intelligence, Decision Support, Corporate Boards, Strategic Foresight, Environmental Scanning, Technology Adoption

Sammanfattning

Styrelser i företag står inför fyra strukturella utmaningar i sitt arbete med strategisk framförhållning. För det första filtreras informationen som når styrelsen av företagsledningen, vilket styr vad styrelsen faktiskt får se. För det andra följer styrelsearbetet en episodisk mötescykel som sällan håller jämna steg med förändringstakten i omvärlden. För det tredje saknas ofta ett systematiskt sätt att bevaka omvärlden utanför den egna branschen. Slutligen kännetecknas scenarioplaneringen av en omedveten optimism, där negativa utfall inte analyseras lika noggrant eller djupt. Artificiell intelligens (AI) framstår som en möjlig lösning, men i dagsläget saknas kunskap om hur AI-baserade verktyg för omvärldsbevakning och analys bör utformas och styras i en styrelsekontext. Syftet med detta examensarbete är att utveckla ett ramverk som möter detta behov.

Studien har Design Science Research som metodisk grund och kombinerar två typer av underlag. Det första är en teorigenomgång inom sju kunskapsområden: begränsad rationalitet, strategisk framtidsanalys, AI i organisatoriskt beslutsfattande, teknikadoption, förtroende och förklaringsbarhet, styrelseprocesser och informationsbearbetning i grupp, samt AI-styrning och förtroendeansvar. Det andra är ett empiriskt material från tio semistrukturerade intervjuer med erfarna styrelseledamöter i nordisk kontext, samt en kompletterande expertintervju om metoder för strategisk framtidsanalys. Materialet analyserades tematiskt och resulterade i åtta teman, fördelade över tre lager (sammanhang, förmåga och användning). Utifrån dessa utvecklades ett designramverk med 21 designprinciper fördelade över fyra dimensioner: kognitiv förstärkning (DP1-DP6), strategisk framtidsanalys (DP7-DP11), förtroende och acceptans (DP12-DP16) samt styrning och ansvar (DP17-DP21).

Resultaten visar att AI har potential att hantera samtliga fyra utmaningar, men endast under tre förutsättningar. För det första måste verktyget utformas så att diskussion och kritiskt tänkande bevaras, eftersom det är detta som gör styrelsearbetet värdefullt. För det andra måste implementeringen av verktyget ske strukturerat, med en tydlig plan och utbildning för ledamöterna. För det tredje måste verktyget styras så att ledamöterna behåller det fulla ansvaret för besluten och inte förlitar sig blint på tekniken.

Examensarbetet ger fyra teoretiska bidrag. Det första är att knyta samman forskningen om strategisk framtidsanalys (corporate foresight) med forskningen om AI i styrelser, med utgångspunkt i en svensk kontext. Det andra är en utvidgning av Forbes och Millikens (1999) processmodell för styrelsebeslut till en AI-kontext. Det tredje vidareutvecklar Raisch och Krakowskis (2021) paradox mellan automatisering och förstärkning (augmentering) genom att tillämpa den på styrelsens särskilda ansvarskrav. Det fjärde är introduktionen av epistemiskt oberoende som en designprincip.

Nyckelord

Artificiell intelligens, Beslutsstöd, Bolagsstyrelser, Strategisk framtidsanalys, Affärsförutseende Omvärldsbevakning, Teknikadoption

Acknowledgments

This thesis would not have been possible without the generosity and engagement of many people. We are deeply grateful to our academic supervisor, Jennie Björk, for consistent and constructive guidance throughout the research process.

We owe particular thanks to our industrial supervisor, Carl Henrik Forzélius at Indea, whose professional network, domain expertise in board governance, and active engagement with the research shaped both the empirical work and the direction of the framework in fundamental ways. Indea's commitment to advancing the quality of board work in Nordic organizations provided the practical grounding that design science research requires.

We are sincerely grateful to the ten interviewees who gave their time and shared their experience with us. Their perspectives, drawn from boardrooms across fintech, energy, technology, pharma and biotech, medtech, and digital health, are the empirical foundation on which this thesis stands.

We are equally grateful to the two experts who contributed to our advisory sessions on foresight methodology and corporate strategy, whose perspectives helped us test and refine our emerging interpretations.

Stockholm, May 2026
Thiyaneshwar Padmanaban & Linda Ronaghi

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List of acronyms and abbreviations

AI	Artificial Intelligence
CEO	Chief Executive Officer
CFO	Chief Financial Officer
CIO	Chief Information Officer
DP	Design Principle
DSR	Design Science Research
ESG	Environmental, Social, Governance
EU AI Act	European Union Artificial Intelligence Act
GDPR	General Data Protection Regulation
RQ	Research Question
SFS	Svenska Författningssamling (The Swedish Code of Statutes)
UTAUT	Unified Theory of Acceptance and Use of Technology

1 Introduction

This chapter introduces the research problem addressed by this thesis, establishes the context and purpose of the study, presents the research questions, and outlines the overall structure of the report.

1.1 Background

In corporate boardrooms, decision-makers are responsible for making strategic choices under conditions of uncertainty. Directors must set long-term strategic direction for organizations, oversee executive leadership, and safeguard the economic interest of stakeholders (The Swedish Corporate Governance Board, 2023). However, even the most competent directors face cognitive constraints when processing vast amounts of data, a behavioral pattern described as bounded rationality. Bounded rationality is the result of time and knowledge limits on decision-makers that cannot be overcome regardless of how skilled the decision-maker may be (Simon, 1955). Decision-making in organizations occurs on the basis of information processed through standardized organizational reporting systems, capable of accounting for all risks known to the organization, yet incapable of processing new risks or weak signals from the external environment (Daft and Weick, 1984). At the same time, the complexity of the strategic environment keeps growing, fueled by innovation, political instability, and rapid change. The success of board decisions depends on information and the quality of information available to board members.

However, the information infrastructure available to corporate boards has not evolved much in the last few decades. Board members typically receive their briefing packages before meetings, which are put together by the chief executive officer (CEO) and the management team (Brennan, Kirwan and Redmond, 2016; Evgeniou *et al.*, 2025). While best practice recommends distributing board packs seven days prior to a meeting, data indicates a common trend of receiving them five working days or less in advance (Pantelides, 2026). Strategy is assessed through the annual strategic review and also through periodic strategy workshops (Evgeniou *et al.*, 2025). Consequently, the type of information presented, and the analytical framing used in such briefings are determined entirely by what management deems relevant for its own operational purposes (Brennan, Kirwan and Redmond, 2016).

This structure may create a gap between the oversight expectations placed on corporate boards and their actual technological capacity to scan the external environment. Strategic disruptions happen more rapidly and frequently than the traditional corporate calendars allow (Haarhaus and Liening, 2020; Marinković *et al.*, 2022). Bounded rationality, management filtering, and the built-in time lag of periodic governance rhythms leave organizations vulnerable to adjacent market threats and blind spots that lie outside their standard scanning (Simon, 1955; Brennan, Kirwan and Redmond, 2016; Evgeniou *et al.*, 2025).

Advanced technology could help address this gap. Artificial intelligence (AI) is able to track large amounts of public information available in real time regulatory statements, competitive disclosures, patent searches, industry publications, macroeconomic figures, scientific articles in scope and in velocity not comparable to what any human foresight team could do (Ahdadou, Aajly and Tahrouch, 2024; Evgeniou *et al.*, 2025; The World Economic Forum and OECD, 2025). Signals can be analyzed across several information categories; patterns identified at the margins of what is familiar to the organization's strategic horizon; and summarized, sourced briefings provided to complement management-prepared information currently available to the board (The World Economic Forum and OECD, 2025; Anwar, 2026).

There is real potential here, but its fulfillment is not merely a question of implementing a competent AI. Corporate board directors are prestigious and accountability-conscious professionals

bound by fiduciary duties to govern through a series of processes that involve collective sense-making, cognitive conflict, and knowledge utilization (Forbes and Milliken, 1999). These processes are something that AI can help or hinder. They are embedded in an environment of uncertainty regarding the accuracy of AI, the security of its data, and its governance that informs trust development in unique ways (Ahdadou, Aajly and Tahrouch, 2025; The World Economic Forum and OECD, 2025; Anwar, 2026).

This thesis has been developed in collaboration with Indea, a company offering board services and corporate governance expertise. Through this partnership, access to interviewees and expert in the field were identified. This research has been conducted in a Nordic context, and more specifically, Swedish corporate governance.

1.2 Problem Statement

Despite growing academic interest in AI for corporate governance, there is a gap between what AI can theoretically do for boards and what board members are realistically prepared to adopt and trust. Existing market solutions for governance are typically limited to generic productivity assistants or administrative board portals focused on document distribution and meeting scheduling. At the same time, the academic literature on corporate foresight identifies a persistent integration problem: foresight intelligence routinely fails to reach the boardroom in a form that directors can act on (Rohrbeck and Gemünden, 2011).

The literature on AI applications in corporate governance has grown rapidly in recent years. Nevertheless, there is a noticeable lack of structural consistency within this literature. There is vast amount of literature on organizational foresight (the structured exploration of multiple plausible futures to inform present strategy) and environmental scanning (the systematic monitoring of the external environment for emerging signals) that considers the theory deeply and proposes an approach that allows organizations to better process strategic information. Yet, most of this literature does not take up the topic from the boardroom angle but rather focuses on operations (Daft and Weick, 1984; Rohrbeck and Gemünden, 2011). In contrast, there is plenty of literature that considers how AI could be used by corporate boards (Ahdadou, Aajly and Tahrouch, 2025; Ben-Shahar *et al.*, 2025) but hardly touches on any foresight theory.

What is missing from the existing literature is a structured design framework that specifies the functional requirements, trust conditions, and governance safeguards needed to successfully deploy an AI-based foresight tool in the corporate board context. This thesis aims to address this gap.

1.3 Purpose

The purpose of this thesis is to develop a design framework for an AI-based decision-support tool that augments strategic foresight and environmental scanning for corporate board. By exploring the gap between theoretical AI capabilities and board members' practical needs, the study aims to create design principles that are theoretically grounded and empirically informed. The intended contribution is a set of design principles that offer guidance on the functional requirements, trust conditions, and governance safeguards that may enable the adoption of AI tools in corporate boardrooms.

1.4 Research Questions

This thesis is guided by the following research questions:

RQ1: What information challenges and foresight gaps do corporate board members experience, and how do these shape the requirements for AI-based decision support?

RQ2: What design and governance principles should govern an AI-based foresight tool for corporate boards to achieve adoption and trust among directors?

1.5 Delimitations

As the research phase began in February, reducing the intended 20-week schedule, the overall scope was restricted to ensure results could be finalized and presented on time. The research is carried out in Sweden, and the participants are therefore limited to the same region. Additionally, participants were required to be either currently active on corporate boards or seniors with significant past board experience. Furthermore, non-profit organizations were excluded, as their boards operate under different objectives and governance logics. Limiting the sample to profit corporate boards keeps participant's context more comparable.

The thesis focuses on strategic foresight and environmental scanning, rather than administrative and compliance-oriented board functions such as meeting administration, regulatory reporting, or performance benchmarking. These are significant and growing areas of AI application in governance, but they raise different design and governance questions which are outside of the scope of the thesis. Furthermore, Annex III of the EU AI Act (*Regulation (EU) 2024/1689 (Artificial Intelligence Act)*, 2024) lists the AI use cases classified as 'high-risk' (e.g. biometrics, critical infrastructure, law enforcement). The framework assumes a board foresight tool falls outside these categories; therefore, the remainder of this thesis addresses general governance principles instead of specific compliance requirements.

Although this thesis addresses specific design criteria in order to make a successful organizational foresight application, building and evaluating a functional prototype remains outside the scope.

1.6 Structure of the thesis

The remainder of this thesis is organized as follows, Chapter 2 presents the theoretical background. Chapter 3 describes the qualitative and design science research design, including literature review process. The results of the study are presented in two chapters: Chapter 4 reports the empirical findings from the interviews, and Chapter 5 presents the resulting design framework and principles alongside their empirical and theoretical findings. Chapter 6 discusses the theoretical and practical implications, research limitations, and future research. Chapter 7 presents the final conclusion of the study.

2 Theoretical Background

This chapter establishes the theoretical foundations on which the analysis in Chapters 4 and 5 builds. The literature is organized to follow the logic of the design problem itself. Section 2.1 begins with the cognitive conditions of board decision-making, followed by an examination of corporate foresight and environmental scanning in Section 2.2. Section 2.3 introduces the literature on AI in organizational decision-making, while Section 2.4 describes the conditions for technology adoption. Section 2.5 reviews trust, verifiability, and explainability. Section 2.6 discusses board dynamics, group information processing, and the risk of AI limiting the deliberation space. Section 2.7 outlines the governance and fiduciary constraints a board-level AI tool must operate within. Finally, Section 2.8 synthesizes these seven streams into the analytical lens used in the rest of the thesis.

2.1 Bounded Rationality and Cognitive Limitations in Decision-Making

Strategic decision-making cannot be modeled as a fully rational process. Real decision-makers operate under bounded rationality, where limits on information, attention, and computational capacity force them to settle for outcomes that are satisfactory rather than optimal, a behavior described as ‘satisficing’ (Simon, 1955). What was originally framed as an individual cognitive constraint has since been extended to the organizational level, where firms have been theorized as interpretation systems in which top management functions as the layer that scans, encodes, and translates external signals into a form the wider organization can act on (Daft and Weick, 1984; Choo, 1994). Bounded rationality is not only a property of individual decision-makers but a structural feature of how organizations cope with complexity, and the design of the interpretive layer determines what information reaches the level at which formal decisions are made. Recent work has extended this argument to the conditions of information abundance, it shows that when the volume of available signals exceeds the cognitive processing capacity, the result leads to inattentive to detail, decision-maker stop engaging with material that exceeds what they can meaningfully process, producing a filtered effect that is different from information scarcity (Ward, 2023).

For corporate boards, these constraints are intensified, as boards meet a limited number of times per year, might work part-time alongside other commitments, and engage with strategic questions of growing complexity while carrying full legal fiduciary responsibility, a combination that has been identified across the literature as the persistent structural condition of board work (Mittelstadt, 2019; Evgeniou *et al.*, 2025). Recent work has named the specific challenges this produces. Information asymmetry between board and management and constrained time between meetings sit closest to the bounded-rationality problem, with additional pressures from group dynamics that are addressed in Section 2.6 (Ben-Shahar *et al.*, 2025; Evgeniou *et al.*, 2025). The board's capacity to act intelligently under uncertainty has been theorized as a function of the heterogeneity of analytical inputs available to it and the diversity of perspectives those inputs sustain, with both individual cognitive limits and collective epistemic capacity treated as interdependent rather than separate problems (Ben-Shahar *et al.*, 2025).

Consequently, boards rely heavily on the CEO and the management-prepared documents that reach them in advance of meetings, and across the relevant literature this reliance is described as a structurally rational satisficing strategy to cope with overwhelming amount of information (Daft and Weick, 1984; Choo, 1994; Evgeniou *et al.*, 2025). However, this dependency causes information asymmetries. Information reaching the board has been filtered twice, by management's scanning preferences and by management's editorial choices, with limited time and access to seek alternative inputs between meetings (Choo, 1994; Brennan, Kirwan and Redmond, 2016; Evgeniou *et al.*, 2025). When epistemic inputs are concentrated in a single channel in this way, the board's capacity for independent challenge gets weakened. The literature describes this risk as 'epistemic capture,'

where the board relies too heavily on management's narrative, leading to a premature agreement on management's preferred direction (Ben-Shahar *et al.*, 2025). To counteract this vulnerability, various authors argue that boards may benefit from analytical support tools to overcome these cognitive limits. They point out that the bounded rationality of decision-makers creates a gap between the formal legal expectations placed on directors and the cognitive resources they actually possess (Mertens, 2023; Zhao, 2024; Evgeniou *et al.*, 2025). This gap, rather than any failure of individual competence, is a structural problem.

Bounded rationality alone, however, does not specify what kind of analytical input would address the deficit. To make the problem tractable, the next section narrows the focus to the specific cognitive task that defines board oversight: scanning the external strategic environment and making sense of what it shows.

2.2 Corporate Foresight and Environmental Scanning

To successfully navigate environmental uncertainty, boards must engage in strategic foresight. The primary function of strategic foresight is to systematically evaluate multiple plausible futures during uncertainty, rather than projecting a single expected outcome over a longer time period (Marinković *et al.*, 2022; Grove, Clouse and Xu, 2023). To be useful in board decision-making, strategic foresight requires more than attention to long-term trends. It requires the ability to think in terms of multiple possible futures, complex interdependencies, and rates of change that exceed linear expectations, which is described as future thinking, systems thinking, and exponential thinking (Grove, Clouse and Xu, 2023). In a corporate context, these capacities are operationalized through two processes. The first is environmental scanning which involves the continuous monitoring of weak signals across technological, political, economic, social, and ecological domains (with horizon scanning referring more specifically to the detection of emerging signals). Second is scenario planning, which structures those signals into alternative futures against which strategic assumptions can be tested (Marinković *et al.*, 2022; Grove, Clouse and Xu, 2023).

2.2.1 Environmental Scanning as Organizational Sense-Making

Whether environmental scanning produces strategic insight depends less on the volume of collected information than on how organizations make sense of it. Organizations are best understood not as information processing systems in the computational sense but as interpretation systems; organizations do not simply receive environmental data, they actively construct interpretations of an external world (Daft and Weick, 1984). Their model identifies four interpretation modes determined by two axes (Figure 2.1).

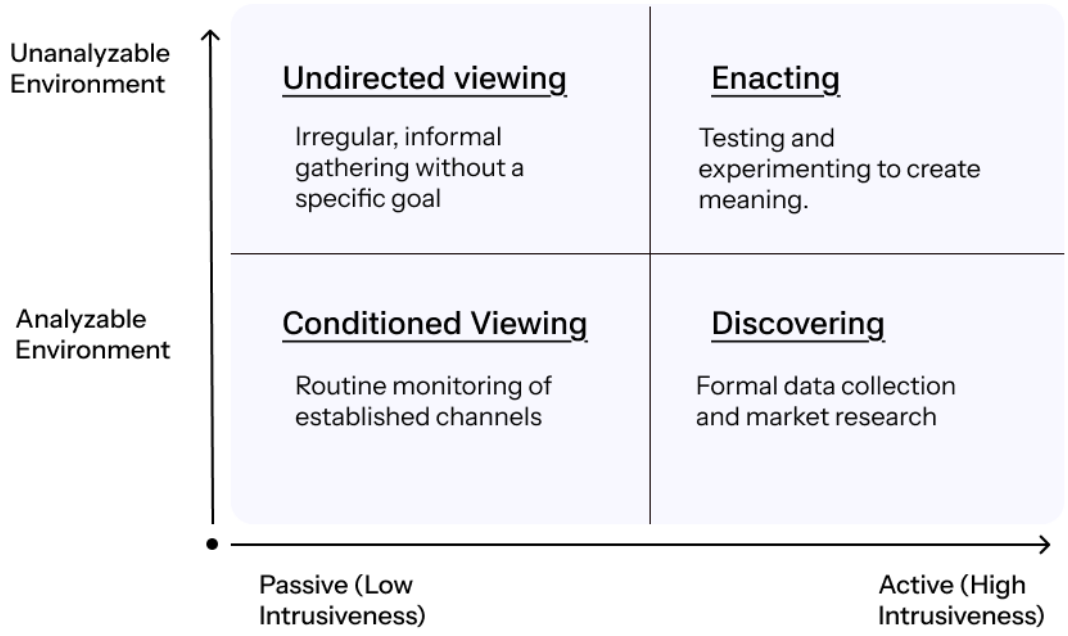


Figure 2.1: **Four interpretation modes for environmental scanning, with two axes: environmental analyzability and organizational intrusiveness, adapted from (Daft and Weick, 1984)**

These two axes represent management's assumptions about the analyzability of the environment, and the organization's intrusiveness (the degree to which it actively probes versus passively monitors). The resulting modes range from undirected viewing (irregular, informal, passive) through conditioned viewing (routine monitoring of established channels) to enacting and discovering (active, intrusive engagement with a complex or analyzable environment) (Daft and Weick, 1984). Studies of how executives actually scan converge on a related point: information is filtered to reduce equivocality, so that the organization attends to what its existing frameworks are already equipped to interpret and discounts what they are not (Choo, 1994; Robinson and Simmons, 2018).

Applied to corporate boards, this typology locates board information practices predominantly within conditioned and undirected viewing. Boards engage with materials assembled through established management channels and covering the domains management has already defined as strategically relevant, which means the environment is monitored rather than actively probed. A recognized consequence of this pattern is the weak detection of peripheral signals, the early and ambiguous indications of disruption that originate outside the currently monitored domain (Daft and Weick, 1984; Grove, Clouse and Xu, 2023). Adjacent industries are a characteristic blind spot, since a competitor converging on the organization's core market from a neighboring sector is unlikely to appear in channels defined around existing competitors. This is not primarily a failure of management capability but a structural property of interpretation: information reaching the board has already passed through the organization's cognitive frameworks and arrives pre-interpreted, classified according to what the organization already knows how to process (Daft and Weick, 1984).

Choo's (1994) study found that senior executives overwhelmingly prefer personal and internal information sources, such as subordinate managers, direct contacts, internal reports over impersonal external sources such as databases and publications. Personal sources were rated higher in perceived reliability and relevance; electronic information services were among the least used sources. The research on this interpreted this preference not as a failure of information literacy but as a rational response to information overload: executives filter by source trustworthiness to

manage the cognitive demand of environmental monitoring (Choo, 1994). The pattern persists in digital information environments, although larger organizations tend to develop more systematic internal processes to compile and aggregate external signals before they reach decision-makers (Robinson and Simmons, 2018).

For boards, these findings imply a structural gap: the information environment boards inhabit is therefore shaped by the cumulative preference and filtering of the CEO and senior management team, who themselves favor personal and internal sources, and whose cognitive frameworks define what is noticeable.

2.2.2 Corporate Foresight as Organizational Capability

Corporate foresight does not only help the organization notice external change; it also helps the organization think more broadly about possible futures and question whether its current assumptions on resource allocations and strategy are still valid. In practice, corporate foresight can be described having three roles (Rohrbeck and Gemünden, 2011); the strategist role explores strategic territory beyond the firm's current frame, catching opportunities and threats that standard planning cycles are not designed to surface. The initiator role expands the pool of strategic alternatives by identifying emerging technologies and market shifts before they become obvious to competitors. The opponent's role uses foresight intelligence to challenge and stress-test existing strategic commitments, not to obstruct management but to subject proposals to scrutiny they might not otherwise receive.

At board level, the opponent's role deserves particular attention because it speaks directly to a recognized weakness in board deliberation. Consensus pressure remains one of the most persistent threats to effective board decision-making; experienced directors may be aware that such pressure should not drive decisions at that level of authority, but boards are still small groups subject to the social dynamics that suppress cognitive conflict (Forbes and Milliken, 1999). The capacity to challenge management's account has been linked to the board's independent analytical access to the environment, with directors found to exercise more meaningful strategic influence when they are not wholly dependent on management's framing of the issues (McNulty and Pettigrew, 1999).

2.2.3 Foresight as Dynamic Capability and the Integration Problem

Strategic foresight is frequently analyzed through the lens of dynamic capabilities. This theoretical framework shifts the focus from a firm's static assets to its capacity to sense, seize, and reconfigure its resources to better respond to environmental changes (Teece, Pisano and Shuen, 1997; Haarhaus and Liening, 2020; Marinković *et al.*, 2022). While this perspective has traditionally been developed with management as the primary point of analysis, recent work has tried to reframe these capabilities at the board level. In this context, "sensing" is defined as the board's independent environmental scanning capability, but something that tends to be neglected (Engstam *et al.*, 2024). This neglect is a result of the overreliance on information curated and filtered by management, which prevents the board from developing an independent perspective on the external environment.

Strategic foresight provides the foundation for this sensing capacity, as it allows organizations to move beyond the limitations of static long-term planning and hierarchical structures. Over time, this capability enables the board to better anticipate change before it results in a crisis (Haarhaus and Liening, 2020; Marinković *et al.*, 2022). Furthermore, strategic foresight has been shown to improve strategic flexibility and decision rationality, providing significant value under conditions of environmental uncertainty (Haarhaus and Liening, 2020). This is particularly relevant in the current governance context, where boards must navigate environments marked by technological disruption, geopolitical volatility, and structural change.

The harder problem, though, is not generating foresight intelligence; it is getting that intelligence into the room where decisions happen. Integration is where foresight practice most often fails. Analytically rigorous outputs routinely fail to reach the board's agenda, absorbed by the organizational and cognitive distance between whoever produced the analysis and whoever needs to act on it (Rohrbeck and Gemünden, 2011). The most demanding part of this process is the synthesis stage, where raw environmental signals must be converted into something a board can actually deliberate on, a translation that places heavy cognitive demands on whoever performs it (Choo, 1994; Li *et al.*, 2018).

The relevant question for board-level design is therefore not whether AI can perform foresight tasks, but how it should be positioned in relation to the human deliberation that follows. The next section turns to the literature on AI in organizational decision-making, which has converged on a specific answer to that question.

2.3 AI in Organizational Decision-Making

Research on AI in organizational decision-making distinguishes between automation, where AI replaces human work, and augmentation, where AI is designed to collaborate with humans rather than substitute them. The two modes are not fully separate as augmentation automates parts of a task (such as signal detection) while leaving interpretation and judgment to humans. For complex and high-stakes decisions, the management and corporate governance literatures converge on augmentation as the more appropriate model, on the grounds that it combines the computational capacity of AI with the contextual judgment, values, and accountability that remain with humans (Daugherty and Wilson, 2018, cited in Raisch and Krakowski, 2021; Ahdadou, Aajly and Tahrouch, 2024, 2025). This is not merely about efficiency, it is also about the ethical position, where AI should extend rather than replace human decision-making (Ahdadou, Aajly and Tahrouch, 2024; Agnese, Arduino and Di Prisco, 2025). The augmentation logic is not only a theoretical position; practitioners have themselves reported treating AI as a complement to human expertise rather than a substitute (The World Economic Forum and OECD, 2025). Although it adds speed and analytical depth, it was noted that it requires careful interpretation by an experienced user. Accordingly, 'AI' is used throughout this thesis to mean artificial intelligence deployed in an augmentation mode, where it assists rather than replaces human work.

Neither automation nor augmentation can be taken as two separate decisions in design. When part of a work process is automated, the surrounding human activities are reshaped into new forms of augmentation: setting criteria for the AI, evaluating its outputs, and deciding when to override them (Raisch and Krakowski, 2021). Attempts to resolve this tension in either direction tend to produce worse outcomes than treating it as a productive complementarity that must be managed rather than eliminated (Raisch and Krakowski, 2021; Ahdadou *et al.*, 2025)

Signal detection involves ongoing monitoring of open-source information from various channels to detect signs of competition, regulation changes, market trends, and new threats and can be automated because of the high level of cognition required to do so, it is clearly defined boundaries, and the capacity of AI to surpass the ability of each individual board member to perform this task comprehensively (Raisch and Krakowski, 2021; Ward, 2023). Signal detection requires making sense of it, determining their strategic implications for the company, and deciding whether the organization's strategic direction needs adjustment, which can be performed only through engagement, as it requires knowledge, values, accountability, and intuitive assessment that are out of the scope of existing AI capabilities (Raisch and Krakowski, 2021; Zeiser, 2024). The engaged augmentation approach outlined by Agnese et al. (2025) is the exact behavior that board members need to exhibit during signal interpretation: They will need to analyze, debate, and integrate all the signals detected by AI into their decision-making process.

Within the corporate governance literature specifically, the augmentation logic in the broad sense has been extended further to articulate a more focused position regarding how AI can be utilized at the board level. In particular, the notion of augmented intelligence, which involves collaboration between humans and AI whereby the latter provides analysis but does not make decisions, appears to have gained consensus among scholars (Ahdadou, Aajly and Tahrouch, 2024; Agnese, Arduino and Di Prisco, 2025; Ahdadou *et al.*, 2025). According to the criteria of five stages of integration: assisted, augmented, amplified, autonomous and autopoietic; augmented intelligence is considered to be a stage that is achievable and desirable, because it retains the responsibility for making decisions within the board rather than transferring it (Ahdadou *et al.*, 2025). This convergence is conditional upon the understanding that augmented intelligence does not operate automatically. The literature identifies two behavioral pitfalls that governance design must actively address: first, over-trust, where directors accept AI output without sufficient scrutiny; and second, algorithm aversion, where directors reject the tool even when it performs better than the existing alternatives (Ahdadou, Aajly and Tahrouch, 2024). The appropriateness of the augmented-intelligence model therefore depends not only on the design of the tool but on governance arrangements that keep board engagement at the productive midpoint between uncritical reliance and unwarranted rejection.

Designing that midpoint presupposes that the tool is used in the first place. Whether directors will adopt a board-level AI tool, and under what conditions, is the subject of the next section.

2.4 Technology Adoption in AI Contexts

Designing an effective AI system does not guarantee that it will be used. Research on technology acceptance shows that adoption depends less on a tool's technical capability than on how users perceive it and the conditions in which they encounter it (Davis, 1989; Venkatesh *et al.*, 2003). The *Technology Acceptance Mode*l identified perceived usefulness and perceived ease of use as the central predictors of adoption intention, with usefulness the stronger of the two (Davis, 1989). The later *Unified Theory of Acceptance and Use of Technology* (UTAUT) extended this view, adding social influence and facilitating conditions: whether peers expect a tool to be used, and whether the surrounding infrastructure supports its use (Venkatesh *et al.*, 2003). A complementary view is the *diffusion of innovations theory* which looks at how use spreads through a group rather than at what shapes individual intent. The theory describes that adoption typically depends on a small number of well-connected, high-status early adopters that serve as the reference for others to follow (Rogers, 2003).

These factors take on a particular character in the Swedish board context, where the typical board member is in their late fifties (PwC, 2024) and serves on a board that decides its own working methods Aktiebolagslagen (2005, SFS 2005:551, Ch. 8, Sec. 6). This means that any adoption of new tool is voluntary at both the collective and the individual level. Under such voluntary conditions, individual judgments of usefulness carry most weight in shaping intention to adopt (Davis, 1989; Venkatesh *et al.*, 2003; Marocco, Barbieri and Talamo, 2024). However, social influence may play an important role during the early stages of a market introduction. A few high-status early adopters can effectively legitimize a new tool, shaping how useful the rest of their peers perceive it to be (Rogers, 2003). For boards, this suggests that broader adoption relies on a few respected directors using the tool first, serving as proof to their peers that the technology is valuable.

Adoption also depends on familiarity with the technology itself. Resistance to AI tends to reduce as users become more familiar with how it works and experience satisfaction with the tool's capabilities (Marocco, Barbieri and Talamo, 2024), and over time the exposure reduces the hesitation that initially discouraged use. The effect of familiarity and experience is more evident in recent adoption data published by World Economic Forum (2025). Among foresight experts who actively used AI, 43 percent rated the technology as essential or highly useful, compared to just 16

percent by those with less experience. The data suggests that without a baseline of familiarity, professionals struggle to grasp the potential and value AI might add to their work (The World Economic Forum and OECD, 2025). This inability to identify the tool's value and potential may be a barrier to effective adoption, as performance-related benefits are a crucial motivating factor to integrate AI into organizational decision-making (Fahnenstich, Rieger and Roesler, 2024).

Adoption is necessary but not always enough; even directors who adopt the tool will not automatically rely on its outputs, especially not during decision-making deliberations unless they trust the technology. The conditions under which trust forms in AI are a substantial body of research and are addressed in the following section.

2.5 Trust, Verifiability, and Explainability

Trust is built on how users experience and adjust to technology over time and across different types of tasks. Research identifies a strict divide between a user's *trust attitude* and their *trust behavior* (Fahnenstich, Rieger and Roesler, 2024). *Trust attitude* refers to the user's stated belief that a system is capable, while *trust behavior* is their willingness to rely on it during a live task. What the user might say about AI and what they do with it are often completely different. For board work this might mean that directors who support such tools in principle, may still revert to traditional preparations method when preparing for a meeting. Once directors begin to use the tool, willingness to rely on it is shaped by their prior expectation of its ability, and less about its objective accuracy (Cooper *et al.*, 2025). Meaning, if the user expects the tool to be unreliable, the information provided by the tool may be discounted for after a minor error, even when its overall performance has proven to be high. Therefore, early outputs carry disproportionate weight in how trust is accumulated over time by the user.

Trust also varies by task type, and when facing high-stake decisions, the human desire to maintain control intensifies, often decreasing a user's willingness to rely on AI (Fahnenstich, Rieger and Roesler, 2024). On fact-driven, knowledge-based tasks, where users have a clear sense of what a correct answer should look like, users tend to disregard AI outputs in favor of their own judgment even when the technology is more accurate (Krügel, Richter and Uhl, 2025). Whereas on moral or normative tasks the tendency can reverse, and users may lean on AI advice to diffuse or offload moral responsibility (Krügel, Richter and Uhl, 2025). In the boardroom context, meetings can span through both task types. Where environmental scanning tends to be closer to the knowledge-based end, strategic direction-setting can converge towards the normative end (Krügel, Richter and Uhl, 2025). This suggests that trust levels might shift during a single meeting in ways that no single design intervention can fully address.

Whether explanation features can correct for these patterns depends on what they explain. The literature on explainable AI separates technical transparency, which is how a model works internally, from explanation tailored to what a particular audience needs in order to act on the output (Barredo Arrieta *et al.*, 2020; Rieger *et al.*, 2025). Those needs can depend on the form, vocabulary, and depth of the explanation, and these must match the cognitive context of the user. For non-specialist decision-makers, the relevant form is described as *rational explanation*, meaning accessible reasoning about why an output was produced (Barredo Arrieta *et al.*, 2020). Additionally, explanation should help the user decide, case by case, when to follow an output and when to look at the reasoning more closely (Rieger *et al.*, 2025). For board members who are experienced in decision-making but non-specialists when it comes to AI, this means that the relevant form of explanation is rational explanation.

Traceability allows the user to follow a claim back to its original source and check the accuracy independently. While serving as a mechanism for verification, this also protects against a risk described as the *attributability gap* (Zeiser, 2024). When AI contributes to a decision, it becomes

harder to determine which judgments belong to the human and which are inherited from the system, weakening the conditions for moral and legal responsibility. Source traceability is therefore a precondition for preserving attributable human authorship, ensuring that hidden assumptions in the AI's analysis do not quietly replace human ownership over the final decision (Zeiser, 2024). However, traceability alone does not protect the user against a more structural risk described as epistemic capture, which takes two forms: *algorithmic groupthink* and *suppression of dissent* (Ben-Shahar *et al.*, 2025). *Algorithmic groupthink* is when biased input or narrow prompts generate outputs that reflect the management preferences while appearing neutral. The second one, *suppression of dissent*, happens when the directors stop questioning the algorithm and accept its output without critically analyzing it. *Algorithmic groupthink* mirrors the loss of *cognitive conflict* problem that Forbes and Milliken (1999) describe, where boards stop challenging each other's views and converge on a single position. The difference is that now it is the AI's outputs that produce this conformity, rather than social pressure within the group. A proposed response to this risk is *epistemic independence* as a governance principle, where boards must actively encourage multiple perspectives for discussions (Ben-Shahar *et al.*, 2025). A possible approach to this is implementing 'Devil's Advocate AI' system, that actively challenges the primary output, which would prevent the board from reaching an agreement based on one output (Rohrbeck and Gemünden, 2011; Ben-Shahar *et al.*, 2025).

Consequently, while traceability and verifiability have the potential to provide independent thinking, if this is not accounted for, an AI tool could create strategic blind spots if it locks the board into a single interpretive frame. Whether trust and epistemic independence can be sustained is not merely dependent on the individual level but also affected by the group's processes where boards make decisions together, which the following section will discuss.

2.6 Board Processes and Group Information Processing

This section addresses the processes through which boards, as decision-making groups, collectively deliberate and reach decisions, and the ways in which AI might threaten as well as support those processes.

Board effectiveness cannot be explained by the composition of a board alone. A board's knowledge, experience, and independence are necessary conditions for sound strategic decisions, but they are translated into decision quality only through the board's working processes, and research on board cognition identifies three such processes as mediating between composition and outcomes: effort norms, cognitive conflict, and the use of knowledge and skills (Forbes and Milliken, 1999; Agnese, Arduino and Di Prisco, 2025). Effort norms refer to the extent to which directors take their role seriously, preparing thoroughly, participating actively, and applying their expertise conscientiously, and they depend on the culture of the board, the expectations set by the chair, and the perceived importance of the issues at hand (Forbes and Milliken, 1999; Pavićević, Haleblian and Keil, 2023). Cognitive conflict refers to substantive disagreement about strategic issues, the interpretation of information, and the direction of the organization, and the group decision-making literature consistently finds that, when properly managed, it raises decision quality by forcing the consideration of alternatives and the testing of assumptions; it must be distinguished from affective conflict, the interpersonal friction that consistently degrades decision quality (Forbes and Milliken, 1999; Ben-Shahar *et al.*, 2025). The use of knowledge and skills refers to the extent to which the full range of competencies present on a board is actually drawn into discussion, which depends both on whether directors are willing to contribute their insight and on whether deliberation is conducted in a way that makes their contributions feel recognized.

Recent work extends this process analysis to argue that boards are not only recipients of environmental information but can themselves become sources of strategic inertia (Agnese, Arduino and Di Prisco, 2025). Preexisting cognitive structures, demographic uniformity among directors,

and the dynamics of interaction within cohesive boards can lead a board to resist acknowledging environmental discontinuities and threats to the organization's current strategic narrative (Forbes and Milliken, 1999; Agnese, Arduino and Di Prisco, 2025; Ben-Shahar *et al.*, 2025). The resulting resistance to change operates at the level of the group rather than the individual: a board may fail to register a strategic threat even when no single director lacks the competence to recognize it, because the processes through which the group deliberates suppress the disagreement and reinterpretation that recognition would require.

This process perspective reframes what an AI foresight tool contributes, and it cuts in two directions. On one hand, the same group processes that produce inertia can be worsened by AI: when a board defers to AI outputs perceived as objective, the heterogeneity of interpretation on which cognitive conflict depends can narrow rather than widen, a risk described as epistemic capture and as a form of algorithmic groupthink (Ben-Shahar *et al.*, 2025). On the other hand, a tool deliberately designed to introduce external stimuli into deliberation, evidence and framings that originate outside the board's existing cognitive structures and social network, can counteract inertia by provoking the substantive disagreement that high-quality strategic decisions require (Forbes and Milliken, 1999; Agnese, Arduino and Di Prisco, 2025; Ben-Shahar *et al.*, 2025).

Whether such a tool can be deployed at all, however, is not settled by its cognitive or deliberative value. Boards operate within a fiduciary and regulatory framework that constrains what an AI system is permitted to do, regardless of how useful it might otherwise be. The final section of this chapter turns to those constraints.

2.7 AI Governance and Fiduciary Accountability

The fiduciary and regulatory constraints that govern board-level AI use operate at a different level from the trust and deliberation processes covered in the previous sections. They determine not how directors interpret outputs but what an AI tool is permitted to produce, who is accountable for its use, and how that use must be documented.

In the Swedish corporate governance code, the board of directors is responsible for the company’s organization and overall direction according to the Swedish governance framework (The Swedish Corporate Governance Board, 2023). This includes tasks such as setting the strategy and oversee the systems used to control operations and risk. While AI may be able to assist in the decision-making process, the board remains fully accountable for the final decisions. Research on board-level AI treat this human accountability as fixed rather than something a new tool can replace (Mertens, 2023; Chriki *et al.*, 2025). This is because AI itself cannot be held liable, as it lacks the legal and professional accountability that govern human decision-making (Mittelstadt, 2019). As a result, responsibility for any AI-informed board decision must remain with the board members.

Since boards hold the responsibility, there are certain limits to what an AI tool potentially can do. The Swedish governance framework is clear that a board cannot give away final responsibility, even when a task is delegated, as the board remains fully liable for the final decision (The Swedish Corporate Governance Board, 2023). As a result, research generally point to augmented intelligence as the only viable model, where AI provides analytical support while the board keeps full decision-making authority (Mertens, 2023; Ahdadou *et al.*, 2025). This augmented approach also aligns with the European Union's AI Act, which places stricter requirements on systems that replace human judgment than on those that merely inform it (*Regulation (EU) 2024/1689 (Artificial Intelligence Act)*, 2024).

The need for human oversight also extends to how the board documents its use of AI. Under the Swedish Companies Act, directors are generally protected from being blamed for poor outcomes, provided they can show that their decisions were made carefully and based on reliable information. If a board relies on AI without clearly recording how the tool was used and assessed, it may struggle

to demonstrate this careful and well-informed approach (Chriki *et al.*, 2025). In a Swedish context, weak documentation could therefore reduce the protection directors would otherwise have: if a company suffers major losses from an AI-influenced decision and no records show how the tool was critically evaluated, it may be difficult for the board to prove that it acted with due care rather than relying on the tool uncritically (*Aktiebolagslagen*, 2005, SFS 2005:551, Ch. 29, Sec. 1). For this reason, keeping a detailed record of how AI informed a decision, aligned with the formal board minutes, is not merely an administrative formality but the board's main way of meeting its burden of proof and protecting its own legal position (*Aktiebolagslagen*, 2005, SFS 2005:551, Ch. 8, Sec. 24).

Beyond documenting how AI is used, responsibility for the tool itself must be clearly placed with specific people. Research on AI risk governance describes oversight as a layered structure, in which operational staff, risk and compliance functions, and independent review each have a defined role, while the board sits at the top with final responsibility for oversight (Schuett, 2025). Within this kind of structure, accountability for a particular tool cannot be left vague. Studies on board-level AI point out that the absence of a named person responsible for the tool is itself a governance risk, since a tool that no one is clearly answerable for cannot be properly overseen (Chriki *et al.*, 2025). As a result, the legitimate use of an AI tool depends not only on how it is designed, but also on a clear assignment of responsibility, both for running the tool and for the board's reliance on its output.

A final governance consideration is regarding the data an AI tool is allowed to access. Responsible AI governance frameworks treat the management of organizational data as a core compliance issue rather than a purely technical one, and warn that improper data processing can carry significant legal consequences (Papagiannidis, Mikalef and Conboy, 2025). In a board setting this creates a specific risk: if confidential strategic information is fed into an AI system, that data leaves the organization's direct control. A respons to mitigate this risk may be by, for example, limiting what the tool is able to process (Chriki *et al.*, 2025). The choice of what data an AI tool can access is therefore a governance decision, one that determines whether the tool can be used without compromising the confidentiality the board is obliged to protect.

Taken together, these conditions reframe what it means for a board-level AI tool to be legitimate. Legitimacy is not established by analytical quality alone but by whether the tool can be deployed within the fiduciary responsibility, bounded AI role, explicit accountability, and controlled data access that the governance literature treats as the conditions of legitimate board decision-making.

2.8 Synthesis: Toward a Theoretical Lens for AI-Based Board Decision Support

The seven theoretical streams reviewed in this chapter converge on a shared problem diagnosis and a set of design constraints that motivate the empirical work of the thesis. The core problem lies in the very nature of boards. They are constrained by bounded rationality (Simon, 1955), depend on the information filtered by management based on its scanning interests and editing (Daft and Weick, 1984; Choo, 1994), and meet on an episodic basis such that continuous monitoring of the environment is not feasible. This produces a situation where boards face a permanent tension between the oversight role, they must play and the capacity to analyze information they possess. As a consequence, it becomes difficult for boards to make use of cognitive conflict and knowledge in their decisions, which leads to poor quality of strategy formulation (Forbes and Milliken, 1999; Agnese, Arduino and Di Prisco, 2025). Integration presents an additional challenge, as although foresight intelligence is successfully gathered, it rarely reaches the boards due to failures in synthesis. This highlights the critical point where environmental scanning comes to an end (Choo, 1994; Rohrbeck and Gemünden, 2011; Li *et al.*, 2018).

An AI-driven decision support system constitutes an adequate solution to the stated problem as the level of augmentation at play is Level 2 of the framework developed by Ahdadou et al. (2025): the tool supplies analytical input, which is then judged, acted on, and held accountable by the board (Raisch and Krakowski, 2021; Mertens, 2023). Signal detection and horizon scanning can be done automatically, but not signal interpretation (Raisch and Krakowski, 2021; Zeiser, 2024). Within this limit, the specific value proposition of the tool would consist of its ability to uncover information and perspectives stemming from sources other than management, thus enhancing the independence of the board's challenge (Forbes and Milliken, 1999; McNulty and Pettigrew, 1999; Rohrbeck and Gemünden, 2011). Yet, there is an equally valid concern here: an output-oriented approach to using such a tool would reduce the number of perspectives considered and thus undermine cognitive conflict, thereby leading to epistemic capture (Ben-Shahar *et al.*, 2025).

Three additional constraints impact the design process. The technology is not adopted based solely on its quality, but rather on its usefulness, social influences, and trust gained through accuracy and verifiability of the sources used by the system (Davis, 1989; Venkatesh *et al.*, 2003; Barredo Arrieta *et al.*, 2020; Marocco, Barbieri and Talamo, 2024). Legitimacy can be achieved only if the system follows the fiduciary obligations in accordance with the corporate governance model of Sweden: limited AI involvement, clear accountability, audited records, and restricted access to data (The Swedish Corporate Governance Board, 2023; Chriki *et al.*, 2025). Successful implementation will require systematic change management, otherwise the tool will be used without a shift in behavior (Marocco, Barbieri and Talamo, 2024).

Four design dimensions emerge from this synthesis, each corresponding to a distinct constraint identified above.

- Dimension 1 - *Cognitive augmentation* requirements specify how the tool must present information to extend board capacity without increasing cognitive burden.
- Dimension 2 - *Foresight function* requirements specify what the tool must scan, surface, and simulate to address the structural foresight and integration gaps.
- Dimension 3 - *Trust and adoption* requirements specify the verification mechanisms, deployment architecture, and change management conditions required for sustained use.
- Dimension 4 - *Governance and accountability* requirements specify how the tool must be designed to preserve the deliberative and legal conditions that make board governance legitimate.

These four dimensions provide the theoretical scaffolding for the interview guide and the analytical framework through which the empirical findings in Chapters 4 and 5 are interpreted.

3 Methodology

This chapter describes the research approach, design, and methods used in this thesis. It details the literature review process, the semi-structured interview study, the thematic analysis method, and the quality criteria applied to ensure trustworthiness of the findings.

3.1 Research Approach

This study adopts a qualitative and interpretivist research approach to investigate how corporate boards acquire and process strategic information. Additionally, it investigates what requirements an AI-based decision-support tool must meet to be trusted and adopted at board level. An interpretivists lens is necessary, since this area of corporate governance involves knowledge that is not quantitatively measurable and depends on how individuals create and experience meaning (Saunders, Lewis and Thornhill, 2016). This lens also makes it possible to explore how board members make sense of their social environment, more specifically, how they view their information sources, decision-making processes, and subjective experience with emerging technologies.

Following Design Science Research principles (Hevner *et al.*, 2004), the primary goal is to provide practical design knowledge for system development. The outcome of the thesis is a framework detailing functional requirements and governance conditions, based on literature and empirical data, rather than a finished software product.

3.2 Research Design

The study was structured in four phases as illustrated in Figure 3.1, each building on the output of the previous one. The first phase focused on a literature synthesis, where existing research on corporate governance, strategic foresight, AI-based decision support, and technology adoption was used to build the theoretical foundation for the study. The specific steps taken during this stage are further described in Section 3.3 regarding the literature review process.

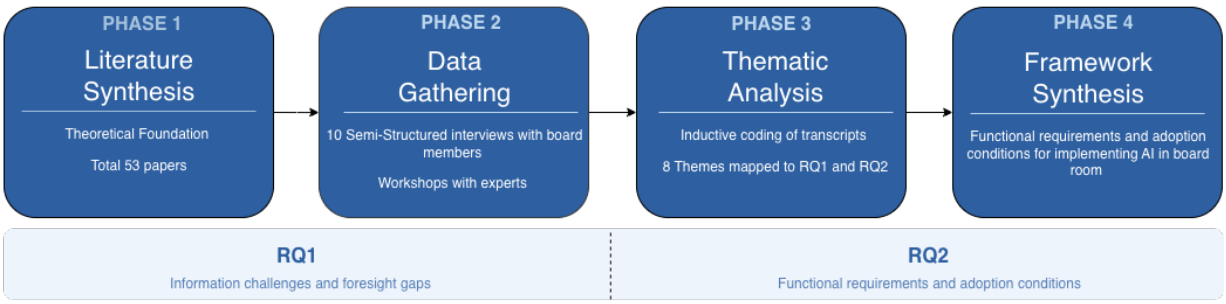


Figure 3.1: Research Design including the four phases

The second phase involved ten semi-structured interviews with board members and one additional interview session with expert to explore and understand the perspective from a practitioner's point of view. The interviews were designed to probe the interviewees' experiences of the information challenges found in Phase 1 and identify requirements, concerns, and design preferences that the literature had not addressed. The methodology for the second phase is further detailed in Section 3.4.

During the third phase, the interviews were transcribed and analyzed using inductive thematic analysis to identify recurring patterns and concerns. a procedure that is thoroughly explained in Section 3.5. Finally, the fourth phase focused on building the framework for an AI-based decision

support tool. The framework synthesis will be based on the theoretical findings from the first phase, combined with the empirical data from the third phase.

3.3 Literature Review Process

The search for relevant literature for theoretical background was conducted using databases such as Scopus, Web of Science, and Google Scholar. The initial search included keywords such as *corporate governance*, *board of directors*, *strategic foresight*, *environmental scanning*, and *AI or artificial intelligence*. To further improve the search results, the KTH Library's AI-assisted search tool, named *Primo Research Assistant* was used to refine and create more precise search terms and strings. By prompting the tool, a summary and five recommended readings were provided; two of these were later used in the theoretical background. The AI-tool also generated a search string that could be used for further search; however, these were too generic, leading to approximately nine million results. To achieve a feasible volume that includes relevant literature, the authors manually refined this string further into three separate combinations (see Appendix A). Additionally, some literature was found by forward and backward citation using ResearchRabbit, an AI-assisted tool that maps citation networks (ResearchRabbit, no date). This approach surfaced research that was of relevance but not captured with keyword search alone.

The primary search phase took place between February and April 2026, with additional searches in May 2026 to include topics from emerging themes found during the interview process. For literature that covered AI, recency and relevance were prioritized over traditional metrics to better cover the current state-of-the-art in AI development, whereas searches covering board work, foresight, governance, bounded rationality, and environmental scanning followed traditional metrics such as journal reputation or citation counts. The recency criterion resulted in inclusion of eight gray-literature and practitioner sources: the World Economic Forum and OECD report on AI in strategic foresight (2025); Working paper by Evgeniou et al. (2025), Mertens (2023), Anwar (2026), Chriki et al. (2025), Engstam et al. (2024); the PwC (2024) survey of Swedish boards, and the Swedish Corporate Governance Code (2023). These were included where they offered relevant empirical or practitioner insight that was not comparably available in peer-reviewed literature. Two primary legal sources, Aktiebolagslagen (SFS 2005:551) and Regulation (EU) 2024/1689, are cited as governing instruments rather than scholarly literature. Remaining literature is from either peer-reviewed journal articles, conference papers, or academic books.

During the primary search phase, the three search strings returned several hundred results, which were screened by title and abstract against the thematic streams of the study. Papers that were not relevant for the theoretical streams, were not in English, or were otherwise outside the scope criteria documented in Appendix B were excluded at this stage. Given the volume and the iterative nature of the search, this screening was guided by established criteria rather than recorded in a formal PRISMA-style tracking protocol, which is consistent with the interpretive and flexible nature of the narrative review approach taken here (Saunders, Lewis and Thornhill, 2016). Papers that passed relevance, based on title and abstract review, together with sources identified through citation tracking in ResearchRabbit, were added into a literature selection map and read in full. The literature mapping initially included 86 papers, where each was classified according to its relevance to the thesis. The classifications were judgement-based and identified as core (foundational for the research questions and directly informing the design framework), supporting (providing broader theoretical or empirical context to one of the streams), or peripheral (background literature referenced where relevant but not central to the argument).

A preliminary structure of six theoretical streams guided the initial search phase: bounded rationality, corporate foresight, AI in organizational decision-making, AI in corporate boards and governance (combined), explainability, and technology adoption. As the interview phase progressed, three refinements were made. First, governance and fiduciary accountability stood out during the

interview process. As a result, a separate section (Section 2.7) was dedicated to addressing the relevant theory, rather than keeping it as part of the broader AI-in-boards heading. Second, trust and verifiability emerged alongside explainability as key concerns in the interview process, leading the original explainability section to be broadened to include all three themes (Section 2.5). Third, a dedicated section on board processes and group information processing (Section 2.6) was added to address how AI affects group deliberation. The supplementary literature searches conducted in May 2026 focused on gathering the necessary sources to support and develop these three refinements. The final reference list ended up with approximately 53 sources, out of which 27 are drawn from the literature selection map and the remainder from these complementary searches.

The approach taken is a narrative literature review, which is an appropriate method for synthesizing literature in qualitative, interpretivist research (Saunders, Lewis and Thornhill, 2016; Sukhera, 2022). A narrative review builds an evolving evidence base in response to emerging themes and the analytical needs of the study (Sukhera, 2022). This flexibility aligns well with the iterative nature of design science research, allowing the researcher to interact with the knowledge base continuously as the study progresses (Hevner et al., 2004). Therefore, this methodology supported the continuous development of the theoretical chapters based on the interview findings, while also allowing for the integration of necessary laws, guidelines, and methodological literature outside of the structured database searches.

3.4 Qualitative Data Collection

The qualitative data was primarily collected through semi-structured interviews with board members; the core interview guide can be found in Appendix C. The guide was developed across three iterations during data collection.

The first version was structured around the theoretical streams identified in Chapter 2, with four to six questions per stream. After the first two interviews, the guide was revised, and questions that had produced generic answers, or where participants struggled to understand, were reformulated. The second version was used for the following five interviews; however, a third revision was done after discussion with both the academic and the industrial supervisor. The second version introduced the proposed tool scenario relatively early in the interview session. The researchers were concerned about the risk of priming the participants, as some participants tended to agree with the presented scenario rather than individually reflect on what they would value. To address these concerns, the third and final version restructured the order of questioning, and the scenario was presented in two stages. The final guide was used for the remaining interviews.

Probes and follow-up questions were prepared and applied flexibly during each interview to avoid further priming. These were used, for example, when an answer remained abstract, and a concrete example would clarify what the participant meant, or when the participant introduced a theme that aligned with one of the research questions.

A standardized introduction and ending script were used by both researchers to make sure participants received the same information before and after the interview. In cases where participants explicitly requested the interview questions in advance, only the core questions were provided, accompanied by a note that the format was semi-structured, and additional follow-ups could occur. In the interviews, 'AI' was used as a general umbrella term without defining the augmentation–automation distinction for participants, so their responses reflect each interviewee's own understanding of AI rather than a predefined mode of deployment. Both researchers attended every session, with one leading the interview while the other took notes and asked follow-up questions that the lead interviewer may have missed. This kept each session balanced between content coverage and depth on unanticipated lines. The leading and supporting roles were rotated

between interview sessions so that both researchers built equal first-hand familiarity with the interview material, rather than one specializing in leading and the other in observing.

3.4.1 Participant Recruitment and Sample

Participants were recruited using a combination of purposive and snowball sampling. This sampling approach was chosen because the target group of corporate board members with active or previously active governance mandates is not easily accessible. Purposive sampling allows the selection of participants to remain relevant for the research questions, while snowball sampling allows for referrals by participants to provide additional interviewees relevant for the study (Saunders, Lewis and Thornhill, 2016). However, both approaches have a risk of producing biased samples, where purposive sampling may reflect the researcher's judgment about who counts as a relevant participant, while snowball sampling tends to result in homogeneity as referrals typically comes from a single professional network (Ahmed, 2024).

To reduce these risks, participants were identified through four independent channels and verified before contact: Targeted outreach via LinkedIn after background checks through Allabolag or comparable public sources (n=4), the researchers' personal network after profile review (n=4), referral from industrial supervisor (n=1), and referrals from participants (n=2). Approximately thirty candidates were contacted across the four channels. Eleven were engaged as participants, two additional candidates accepted but withdrew before their scheduled session, and the rest either declined or did not respond within the data collection window. The variable response rate reflects the difficulty of accessing senior governance professionals, who are typically time-constrained and not part of any standard sampling frame. Data collection was bounded by the thesis timeframe rather than by formal saturation; the final interviews showed few signs of new content emerging, though full saturation cannot be confirmed.

Beyond these channel choices, the only formal inclusion criterion was current or recent board membership at a Swedish or Nordic company, with at least five years of board experience to address the research questions. Participants who held specialized expertise relevant for the research questions were exempt from this criterion due to their unique insights. No restrictions were placed on sector, company size, ownership structure, or AI familiarity, to allow for variation in sample. Previous rather than active board engagement was accepted where the participant held extensive and relevant board experience. Board engagement in startups was excluded to focus the sample, although they would have been considered if the final sample had proven insufficient. Age and gender diversity was sought but not prioritized over relevance to the research questions.

The recruitment resulted in a sample spanning seven sectors: Fintech, Energy, Technology (including Tech and B2B Tech/Digital), Pharma/Biotech (including Pharma and Pharma/Biotech), MedTech, Digital Health, and Foresight Methodology. Ownership structures included listed, private, and private equity-backed organizations, with company sizes ranging from mid-size to large corporations. Collectively, the participants had served on a wide range of boards across their careers, with individual participants having held between one and twenty-five board mandates, reflecting extensive collective governance experience. AI familiarity ranged from low to high (specialist). The scale comprises four levels: low, indicating limited interaction with AI; moderate, indicating regular use in both professional and personal settings but not to its full potential; high (active user); and high (specialist). The distinction between active user and specialist is relative to the board context, where active user refers to someone who regularly uses AI tools and specialist to someone with expert-level AI knowledge in specifically the board context. Five participants were women, including the foresight specialist, and six were men. A full participant profile is provided in Table 3.1, where participants are identified by role and sector to protect confidentiality, and only role and/or background information is used for citations.

Table 3.1 **Interview sample overview**

Informant	Role / Background	Sector	Board Experience	AI Familiarity
I1	Fintech Chairman	Fintech	10+ years	High (active user)
I2	Board Member	Energy	7 years	Moderate
I3	Former Group CEO	B2B Tech/Digital	20+ years	Moderate
I4	Former Head of R&D, International board member	Pharma / Biotech	30+ years	Low
I5	Former CFO	Tech	9 years	High (active user)
I6	Former CEO	MedTech	20+ years	Moderate
I7	Former CEO / Chairman	Pharma	20+ years	Low
I8	Board Member	Fintech	2 years	High (active user)
I9	Head of Innovation	Digital Health	10 years	High (specialist)
I10	Professional board member	Tech	20+ years	High (active user)
I11	Expert Advisory Session	Foresight Methodology	N/A (expert advisory)	High (specialist)

Prior each session participants received information detailing the study's purpose, data handling procedures, and their right to withdraw at any time. Sessions were conducted in English between February and April 2026, primarily via video call and with two conducted in person. Interview durations ranged from forty-five to seventy-five minutes for the regular interviews and thirty minutes for the expert session. With informed consent from the participants, all sessions were audio-recorded and transcribed in full using automated transcription software with manual correction to ensure accuracy before analysis.

3.4.2 Advisory Workshops

In addition to the semi-structured interviews, two advisory workshops were held to support the analysis. The first was an expert advisory session on foresight methodology, and the second was a session with an owner of Indea specialized in corporate strategy. Both sessions took place during the analysis phase and were used as a supplementary input rather than as primary interview data. Their purpose was to test ideas, assumptions, and emerging interpretations, at times grounded in the interview data and at times in theory and prior literature, in order to broaden the researchers' perspective and guide the synthesis of findings. Both sessions were documented through notes, supplemented by audio recording.

The expert advisory session was originally scheduled as a one-hour interview using the same guide as the other sessions. Following coordination with the industrial supervisor, who engaged the

same participant for a separate brainstorming workshop, the session was reduced to thirty minutes. As a result, only a subset of the interview guide was used, and the conversation focused on foresight methodology rather than the full set of research questions. The session is therefore treated as an expert advisory contribution that informed the analysis rather than as a comparable interview. Their participant profile is included in Table 3.1 as *Expert Advisory Session (I11)*.

The second workshop was an online session lasting approximately seventy minutes, held with an owner of Indea with expertise in corporate strategy. The discussion covered how companies work with strategy, how strategy work can become more effective, and how boards can contribute more to strategy, including common pitfalls. The conversation then turned to several of the themes that had begun to emerge from the interviews, which were discussed and tested for plausibility. The session was used to challenge and refine the researchers' interpretations of the interview data and the emerging design framework. Since this workshop served to inform interpretation rather than to generate primary data, their input is described in this chapter and is not coded or reported separately in the findings.

3.5 Data Analysis: Thematic Analysis

The transcripts were analyzed using thematic analysis following the six-phase approach developed by Braun and Clarke (2006). The approach was primarily inductive, in that codes and themes were developed from the interview data rather than from a predetermined coding scheme. The interview guide was, however, informed by the seven theoretical streams reviewed in Chapter 2, meaning the analysis was theoretically oriented even where the coding itself was data driven. This combination was appropriate for exploring board members' subjective experience of information challenges and AI adoption within the theoretical scope recognized by the literature.

In the first phase, both researchers familiarized themselves independently with the transcripts, reading each interview in full and noting initial impressions of recurring topics. In the second phase, each researcher developed an individual set of preliminary codes and candidate themes. The two sets were then compared and discussed, allowing convergent observations to be confirmed and divergent ones to be examined.

In the third phase, the transcripts were printed and physically cut into coded excerpts, which were sorted and re-sorted on a shared workspace until the codes clustered into more concrete theme groupings. This physical sorting allowed iterative grouping and regrouping in a way that supported visual comparison across interviews.

In the fourth and fifth phases, where themes are reviewed, defined, and named, Claude was used as a sense-checking support. The tool was prompted to review the proposed theme names, to assess whether the codes within each grouping appeared logical, and to indicate whether any alternative themes could be identified. No additional themes emerged beyond those already identified by the researchers. However, since the AI tool did not have access to the underlying interview material and could not interpret meaning beyond the labels and quotes provided to it, this result should be interpreted as a directional check rather than as confirmation that the theme set is absolute. The AI tool also suggested presenting the themes in a layered structure of Context, Capability, and Adoption, which the researchers reviewed against the data and found to be a suitable structure for the empirical findings.

In the sixth phase, the themes were written up into the structure presented in Chapter 4. Quotations from participants were selected to illustrate the empirical grounding of each theme. The full thematic coding output, including thesis themes, code labels, descriptions, interviewee reference IDs, and discussion area is provided in Appendix D.

3.6 Research Quality

The quality of the thesis is assessed against Lincoln and Guba's (1985) four criteria of trustworthiness: credibility, transferability, dependability, and confirmability. These criteria are widely used in qualitative methodology (Korstjens and Moser, 2018) and each criterion is addressed in separate sections below.

3.6.1 Credibility

Credibility addresses the confidence that can be put in the truth of the findings and the interpretations drawn from them, meaning the question of whether the analysis genuinely reflects the perspectives of the participants (Lincoln and Guba, 1985, pp. 289–331). Lincoln and Guba propose several techniques for establishing credibility, of which source triangulation, investigator triangulation, and member checks were applied in this study.

Source triangulation was sought through drawing on multiple categories of participants. Ten semi-structured interviews with board members provided the primary data, and one expert interview with a specialist in foresight methodology offered an additional source against which the interview findings could be compared. Where convergence emerged across these sources, the interpretation could be advanced with greater confidence; where divergence arose, it was retained as a substantive finding.

Investigator triangulation was pursued through the involvement of two researchers throughout the study. Both researchers attended all interview sessions, with one conducting the interview and the other taking notes and asking follow-up questions. During the initial phase of the thematic analysis, each researcher independently familiarized themselves with the transcripts and developed independent sets of preliminary codes. The two sets were then compared, with convergent and divergent interpretations discussed before the final themes were defined.

Member checking is described by Lincoln and Guba (1985) as the most essential technique for credibility. After initial data collection and synthesis, summaries were sent back to the interviewees to confirm the accuracy of the data, allowing them to correct errors of fact or to challenge interpretations that did not match what they had intended to convey. This step was intended to support the goal that the findings reflect what participants meant to convey rather than the researchers' interpretations. Seven of the participants responded within the given timeframe. Among these, no changes were requested to the quoted material or to the content of the summaries, and one participant revised their participant profile description. The remaining three did not respond. Given the time constraints of the thesis and the limited availability of board directors, the summaries specified that ‘a lack of reply within the given timeframe would be interpreted as a confirmation’. Comparing the non-respondent’s contribution against the confirmed summaries, there were no major differences in theme coverage or analysis made by the authors.

3.6.2 Transferability

Transferability refers to whether the findings of a qualitative study may be relevant in other settings. The responsibility of the researcher is not to claim broad generalizability but to provide *thick description* of the participants and the research process so that readers can judge the applicability of the findings to their own context. The transferability judgment is therefore made by the reader, not by the researcher (Lincoln and Guba, 1985, pp. 289–331; Korstjens and Moser, 2018).

Thick description was provided across several dimensions in this thesis. Delimitations are described in Section 1.5, where the primarily Swedish governance context of the research is set out.

Secondly, the sample is documented in Table 3.1 in Section 3.4.1 by role, sector, board experience, and AI familiarity, with eleven sessions in total. The same section further discusses gender distribution, sampling strategy, recruitment channels, and the inclusion and exclusion criteria. The interview procedure, including the standardized introduction and ending script, the session format, the language used, and the session duration, is also described in Section 3.4.1, with the full interview guide available in Appendix C. Together, these provide the *thick description* which readers may form their own transferability judgments about whether and how the findings apply to their own contexts.

3.6.3 Dependability

Dependability refers to whether the research process is logical, traceable, and clearly documented, such that another researcher could follow the steps that produced the findings (Lincoln and Guba, 1985, pp. 289–331; Korstjens and Moser, 2018). Lincoln and Guba (1985) propose the inquiry audit as the primary technique allowing for dependability, where an external reviewer examines both the process of the study and the audit trail materials it produced, including raw data, analytical records, and methodological decisions.

Given the scope and timeframe of this study, no external audit was conducted. Instead, documentation of the research process and maintenance of audit trail materials formed the basis for dependability in this study. The documented process includes the procedural steps described under transferability (Section 3.6.2) and the methodological process described in Chapter 3. In addition, the raw data used as a base for the analysis was saved, including audio recording and full transcription. Together, these materials provide the components of an audit trail in the sense described by Lincoln and Guba (1985).

Throughout the study, the academic and industrial supervisor provided an additional layer of methodological review. Practical decisions were discussed continuously during supervisor meetings, including the formulation of the interview guide, the recruitment approach, and the analytical structure. Where uncertainty arose, such as concerns about whether specific interview questions risked being led, the questions were revised with supervisors before further use. In addition, the two advisory workshops described in Section 3.4.2 contributed a further external perspective during the analysis phase. By testing emerging interpretations against the perspectives of the experts, the workshops functioned as a form of external check on the researchers' reasoning. This supervisory and advisory engagement does not represent peer debriefing or an inquiry audit in the strict Lincoln and Guba (1985) sense, but it provided continuous external review of the research process at a level appropriate to the scope of the thesis.

3.6.4 Confirmability

Confirmability refers to whether the findings reflect the data provided by participants, instead of reflecting the researchers' preconceptions (Lincoln and Guba, 1985, pp. 289-331).

Confirmability in this study is supported through the traceability built into Chapters 3 and 4. Chapter 3 documents the process and analytical steps taken, including the considerations behind them and how the final decisions were reached, so that any analytical claim can in principle be traced back to its origin. The findings presented in Chapter 4 are grounded throughout in direct participant quotations, which are intended to provide the visible link between the analytical claim and the source data.

Additionally, the investigator triangulation and member checks described in Section 3.6.1 also support confirmability by reducing researcher bias and giving participants opportunity to challenge interpretations. Section 3.5 and 3.7 include transparent reporting of AI usage, including its

limitations, to allow the reader to assess where the analytical process was supported by tools and where it was driven by the researchers' own judgment.

The iterative refinement of the interview guide, described in Section 3.4, may also have contributed to confirmability. The third revision of the guide reordered the questioning so that participants had the opportunity to give their own reflections on future strategic work and the role of AI before any proposed tool scenario was introduced. This revision was done to reduce the risk of data becoming a reflection of the researchers' framing of the tool rather than what participants independently identified as relevant.

While no formal reflexive journal was maintained in the sense Lincoln and Guba describe, the continuous supervisor discussions noted in Section 3.6.3 provided space for some form of reflexivity: critical self-reflection on the researchers' biases and preconceptions during analysis and interpretation (Korstjens and Moser, 2018). For example, the industrial supervisor offered alternative interpretations of emerging findings, challenging conclusions that appeared to form too quickly; the academic supervisor pointed out the value of transparent reporting even where expected patterns did not emerge.

Taken together, these approaches support the conditions for confirmability, leaving the final judgment to the reader.

3.7 Use of AI

This thesis made use of AI-assisted tools at several stages of the research process. The use is reported here in the interest of transparency, given both the relevance of AI use to the thesis topic and the explicit disclosure.

During the literature search phase, the KTH Library's AI search tool was used to refine search terms, and ResearchRabbit was used to map citation networks and identify additional relevant literature. Claude was used to organize identified research in a working spreadsheet, which helped maintain an overview as the volume of literature grew. All sources cited in this thesis were read, evaluated, and selected by the authors.

During the thematic analysis, AI was used as a sense-checking aid to review proposed theme names, assess the internal coherence of code groupings, and suggest the layered structure later adopted in Chapter 4. The full account of how AI was used at this stage, including its limitations, is given in Section 3.5. All analytical interpretations and theme definitions are the researchers' own.

During the writing phase, Claude and Gemini were used to support consistency of tone and phrasing across sections as the content was written by two authors. The tools were used for revision support and not for generating new content. Example of typical prompts included instructions such as "*check for spelling and errors without changing the argument*", "*suggest edits to bring the tone closer to the surrounding text*", and "*this text feels repetitive and long, identify redundancies and suggest how to improve it*". Any AI-suggested edit was manually reviewed and adjusted to preserve the original meaning and to keep the authors' own voice, especially where content was shortened or restructured.

All theoretical arguments, design decisions, and analytical interpretations presented in this thesis are the authors' own. AI was used to support, not to replace, the analytical and writing work. Before accepting any output, the authors verified it against their own understanding of the material and manually rewrote it as needed.

3.8 Ethical Considerations

Informed consent was obtained from all participants prior to each interview. Participants were informed of the purpose of the research, the intended use of the data, and their right to withdraw at any point without consequence. Participation was entirely voluntary. No participant was compensated and no professional relationship was conditional on participation.

All participants were anonymized in the analysis and reporting. Participants are identified in the thesis by role and sector rather than by name, and organizations are not identified. Role and sector descriptors were chosen at a level of generality intended to limit identifiability, although complete anonymity cannot be guaranteed in studies involving senior professionals in a relatively small national governance community. Where combination of role and sector might have been identifying for a certain participant, further generalization of description was done.

The thesis was conducted in collaboration with Indea, which provided an industrial supervisor with domain expertise. To preserve research independence, the design of the interview guide, the analysis of data, and the conclusions presented were done excluding the industrial supervisor. Interview recordings and transcripts were stored securely on password-protected institutional storage, accessible only to the two researchers. Additionally, the industrial supervisor was not participating in any of the interviews and did not receive access to interview recordings or transcripts.

4 Empirical Findings

This chapter presents the empirical findings presented in a three-layered framework. By combining interview data with existing theory, the following sections identify the capabilities and governance conditions required for a board-level AI tool.

4.1 Overview and Structure of Framework

The codes that were identified during the analysis were organized into eight thematic categories. These categories were further clustered into three layers, as illustrated in Figure 4.1, with each layer addressing a different element of board governance work.

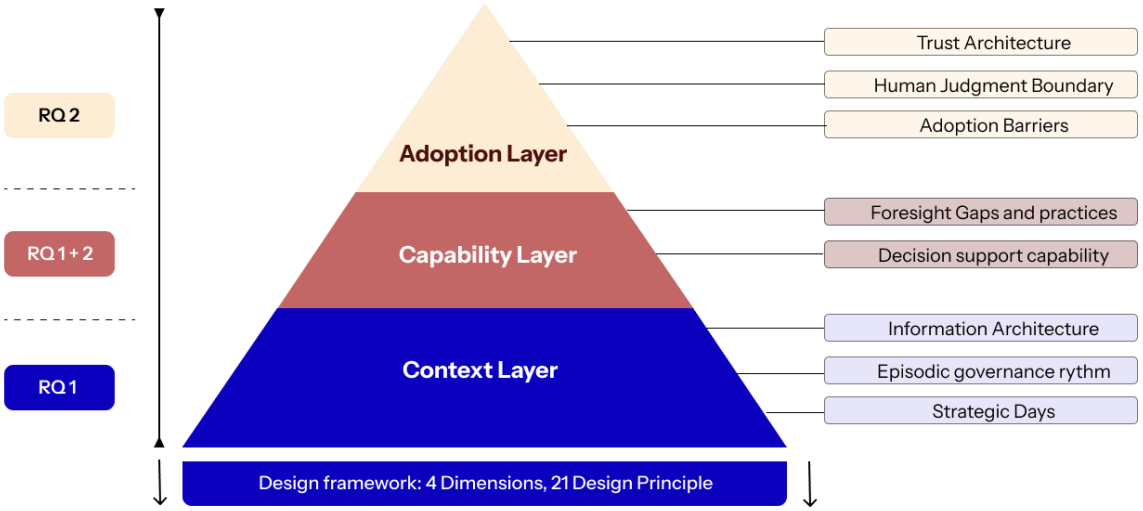


Figure 4.1: **Three-layer empirical framework derived from the thematic analysis, organized into the context layer, the capability layer, and the adoption layer.**

The **Context Layer** contains themes one through three, which describe the structural conditions governing how boards currently access and process information. The themes are the information architecture that boards work within, the episodic rhythm of formal governance, and the strategic days that serve as the primary integration point for forward-looking discussion.

The **Capability Layer** includes themes four and five, which explain what board members have identified as lacking or insufficient in current practice. These themes are the foresight activities boards engage in, and the decision support capabilities they characterized as necessary but not always available.

Finally, the **Adoption Layer** includes themes six through eight, which specifies the conditions which board members would be willing to use AI as external analytical support. These themes are the trust conditions required, the boundaries of human judgment they regard as non-negotiable, and the organizational conditions that need to be in place for adoption to take place.

Additionally, the traceability and relation between the empirical findings (Chapter 4) and the design principles (Chapter 5) is further described in upcoming Section 5.1.1.

4.2 The Context Layer

The three themes in this layer together describe the structural circumstances that limit a board's access to forward-looking external analysis, a problem that exists irrespectively of how well-prepared individual board member is. Theme 1 describes how information is curated and summarized before reaching the board. Theme 2 examines how the frequency of formal meetings influences the continuous strategic oversight, the board's ongoing supervision of the company's strategic direction. Finally, Theme 3 focuses on strategic day(s), which are the dedicated time in the board calendar where long-term strategic discussions are mostly concentrated. Together, these themes explain some of the barriers for effective information processing at the board level.

4.2.1 Theme 1: The Information Architecture Problem

A theme identified from the interviews is that board level information is not a shortage of data, but rather a consequence of how information is filtered, processed, and delivered before a board meeting. Three structural features describe this architecture: the editorial control by the chief executive officer (CEO), the format and volume of material, and the uneven supplementation that boards rely on to compensate for the limits of the first two. The features are interconnected, and together they shape the analytical foundation on which strategic decisions are made.

The CEO as the editorial control

Almost all interviewees described that the board information is directly linked to materials that are produced by the management team and later off confirmed and signed by the CEO, before reaching the board members. Some interviewees describe the material comes from the CEO, while others described it as different internal departments curating the material and that "the CEO will always have the last say about material that goes out", with financial sections being prepared by the CFO and strategic sections, in larger organizations, being drafted by a dedicated strategy or business development function. However, all interviewees agreed that the material needed to go through the CEO to avoid surprises in the boardroom, since the outcome would be time consuming otherwise.

> "*Nothing should come up to the board without the CEO's knowledge*"- Fintech Chairman (I1)

This approach serves as a coordination function for information flow, as it avoids executives being blindsided during board meetings and at the same time provides a single point of editorial accountability for the board material. However, the same approach creates a structural dependency as the board's information is directly relied upon the CEO's capacity, attention, and judgment of the material. Some of the interviewees note that this is more common for smaller organizations, where there is a constraint in recourses. While some describe that the CEO are not specialists in all the strategic areas and therefore miss opportunities or overlook risks. This dependency can be described as information asymmetry between boards and the executives, as the material determines not only what the board sees but also how it is framed.

Material volume, format, and timing

A second feature is regarding the materials itself, as almost all interviewees describe receiving extensive pre-read documents in advance of board meetings. Predominantly these are presented in slide deck format, with some written without capturing the information the board members are seeking. The volume of material varied widely across the interviewees and the organizations they work with, with typical decks of 70 to 80 pages and exceptional cases reaching 400 to 500 slides described as "terrible".

> *"The main problem is not a lack of external information, because that I can always get [myself]... If there is any lack or deficit, that would be in how management describes the issues at hand or their capability of analyzing and then expressing that analysis in words"* - Professional Board Member, Tech (I10)

The challenge the board is facing is not regarding limited material but rather understanding the analysis the management is trying to put forth. External data is broadly retrievable, but the synthesis of that data lives within the management team and their capability to communicate this through the materials they have produced.

When it comes to timing, most interviewees receive materials a week in advance as a matter of policy, though valid exceptions exist, such as waiting for month-end financial data to ensure accuracy. However, a recurrent issue, particularly in less mature organizations, is inconsistent with delivery. Materials often arrive late or in "bits and pieces," with one extreme case leaving the board with only an hour to read the information before the meeting, removing any opportunity for independent preparation

Uneven complementary individual search

The third feature is the independent research that board members do to compensate for the material they have received from the management. All interviewees received the same question, whether the board material was sufficient and if not, how will they act. Several interviewees explained using both traditional search tools and AI tools such as ChatGPT, Copilot, perplexity etcetera when topics fell outside their prior experience. *Professional Board Member*, *Tech (I10)* described looking into external sources when board material referenced a specific competitor or potential partner. However, the interviewee later described that the time spent on such research was estimated to take approximately five percentage of their preparation time, while the majority of the time was spent on reading the board material. Similarly, one interviewee described seeking "information somewhere else" when they did not understand or wanted deeper insight on a topic (I6). *Fintech Chairman (I1)* adopted a more actionable approach, stating that they would "hunt the CEO down" when material was missing. On the other hand, one interviewee explained that they would not necessarily ask the management for clarification before a meeting, "as they probably are very busy and stressed the days before the meeting" (I3).

Beyond reactive verification, several interviewees described a more proactive supplementation pattern, where doing their own research to complement official board materials and to create their own opinion on the topics was crucial. This could, for example, be done by monitoring sector developments, reading industry reports, and searching outside the official information channels.

> *"I didn't just read the information I got but also asked questions which I felt they didn't really address. I could go and check, really to find more [information] and also to give a different complementary perspective. I would say many times the background material was excellent, but equally as often, there was clear gaps in it"* - Former Head of R&D, Pharma/Biotech (I4)

> *"As a board member you want to do your own research. That could be depending on the topic, and it might be very limited, but at least try to go your get your own opinion"* - Former CFO, Tech (I5)

This behavior partially compensates for lack of information in the material that the board receives. However, such behavior is directly linked to individual board members, example given, how much time they have to spare for individual research, their networks and previous experience, analytical habits, desire of arriving well prepared, and desire of not relying solely on management material.

The interview data portrays this well; some are actively seeking complementing data, while others do not see the necessity. The variation was not explained by seniority or experience since both highly experienced and less experienced board members fell on both sides of the divide. This may imply that the analytical foundation entering the boardroom is not uniformed across members of a board.

4.2.2 Theme 2: The Episodic Governing Trap

The rhythm of board governance is structurally episodic, where boards meet a small number of times per year for regular meetings, which the interviewees reinforced. A majority of the interviewees also described that long-term planning was concentrated on one or two dedicated strategic days done together with the management. These rhythms create a strategic stability while ensuring long-term decision-making is addressed, but at the same time it leaves the board and the organization exposed to environmental shifts. This is described here as an episodic governance trap, where the structure is intended to create stability, but instead might allow for gaps in strategic decision-making between these annual- and biannual meetings.

The annual cycle of strategy

Most interviewees described an annual rhythm in which strategic discussion is concentrated into one or two dedicated days, with the regular board meetings between handling business planning, financial review, and operational oversight. The strategic day functions as the main venue for forward-looking discussion, and its content and structure are examined in detail in Theme 3 (Section 4.2.3).

The format itself is described in various ways by the interviewees. Some companies hold dedicated strategic days, while others distribute strategic work across the year supported by a continuous corporate strategy function within management (I2). At the other end of the scale, larger organizations may maintain multi-year strategic plans with excessive analysis and work put into it (I4).

> *"Every company needs its own strategic ways … not a one size T shirt that fits all"* - Board Member, Energy (I2)

What is preserved across these variations is the structurally episodic character of the board's strategic engagement. Alternative formats do not result in more frequent board contacts but rather replaced with increased management work.

Disagreement on the rhythm

The interviewees disagreed on whether the existing rhythm should be changed or altered to adapt to rapid external shifts, and the disagreement itself is a finding. Several interviewees argued that the current pace for long-term planning and strategy needs to stay stable, and revisiting the strategy more often would not automatically add value.

> *"Strategy doesn't need to be continuous challenge… you don't change a strategy very often. It's even if it's important sometimes to do that. But then something is wrong [with the strategy to begin with]"* - Former CEO, MedTech (I6).

Board member, Energy reinforced that, agreeing that the desire is to not change long-term plans frequently, especially in capital-intensive industries. At the same time, *Board member, Energy* adds that changes are necessary to address the dynamic world we are living in right now.

> *"That's kind of the paradox with strategy or these long-term topics. You want it to be something that you don't change all the time. But on the other hand, things are happening all the time."* -Board Member, Energy (I2)

Other interviewees questioned whether the existing rhythm is too slow for the current environment and whether it is "the traditional way of working" standing in the way. *Head of Innovation, Digital Health (I9)* expressed that the traditional rhythm of meeting quarterly or yearly to review strategy is "too late" to adjust strategic direction in response to ongoing external events.

> *"One millisecond after you have endorsed it... it's obsolete"* - Fintech Chairman (I1)

Fintech Chairman (I1) reinforces *Head of Innovation, Digital Health (I9)* concern by arguing that planning for a certain topic might age immediately because the world it is described in does not stop moving. *Former CEO/Chairman, Pharma (I7)* thoughts can be placed somewhere in between, noting that organizations are supposed to revisit strategy annually, but then adds that

> *"Quite often, I would say something disruptive happens that you have to act, and you can to some extent rely on what you have in your strategic plan, but sometimes you probably need to find out where you are, what to do now"* - Former CEO/Chairman, Pharma (I7)

What this disagreement reveals are that the rhythm of board work is challenged even among experienced directors, and that the structurally episodic nature of formal contact might not be the best practice in a dynamic world.

4.2.3 Theme 3: Strategic Days as an integration point into the Board structure

While Theme 2 examined the formal rhythm of board governance and it being structurally episodic, Theme 3 investigates the dedicated setting in which long-term strategic discussion is mainly done. The interviewees described strategic days as the primary integration point of the governance year, where management analysis and boards judgments are brought together into discussion which the ordinary meetings do not address to the same extent.

The interviewees described these days differing from regular board meetings in both length and focus. While regular board meetings are mainly organized around maintaining and updating, strategic days have a forward-looking approach. *Former CFO, Tech (I5)* described spending two days "looking at... external factors and competition... and what choices do we have ahead", followed by "execution plans and what our financial targets are". *Professional Board Member, Tech (I10)* described well-functioning strategic days as collaborative occasions where the board and management identify "what our main forks in the road are that we need to discuss and then form ideas on how to manage them".

Several interviewees describe the strategic day(s) as a multi-stage progression, where material is prepared in advance, discussions between management and board during the offsite, and if the board does not agree with the management, the discussion will have to be adjourned until the issue is revised. The strategic day is therefore not a single point of event, but distributed across the weeks before, during, and after the offsite itself.

Beyond their forward-looking format, strategic days function as a sequenced process, where the focus starts at addressing external signals which then funnels down to the internal perspective.

> *"You go through all macro-events, what is the outlook? See opportunities, cracks in the market, any competitive movements. So, you do the funnel from external things, and then you would funnel it down into internal things. And go through your SWOT analysis [etc.]"* - Former Group CEO, B2B Tech/Digital (I3)

At the very top of that strategic funnel is a clear need to look outward. *Expert Advisory (I11)* addressed this, arguing that boards "really need to understand the macro environment, not just the organization, or even the competitors or just the market". She added that boards "quite often" stop at the industry or sector level when a strategic question requires a broader macro perspective. *Former Head of R&D, Pharma/Biotech (I4)* confirmed this pattern from his own experience from

organizations of varying sizes, pointing out that smaller companies in particular tend to be "very narrow in their field", focusing solely on direct competitors rather than the broader landscape.

To counter this, the interviewees described bringing in outside perspectives directly into the discussion to disrupt the “narrow” point of view.

> *“Not seldomly, you ask someone completely… external. For instance, students and customers to come and provide your insight of what's happening in the world”* - Fintech Chairman (I1)

By implementing these practices, they actively attempt to broaden the perspective and discussions during the strategy day. These external voices can help the board expand their perspectives while aligning their focus to the world beyond the organizational wall. This, in turn, can help ensure that long-term decisions are based on a more comprehensive understanding of the environment outside the company, rather than relying solely on internal operational conditions.

4.3 The Capability Layer

The two themes in this layer emerged from the interviewees describing both the foresight activities they already engage in and the capabilities they identified as limited or absent. Theme 4 examines current foresight practices and the gaps recognized in them. Theme 5 turns to the specific functional requirements board members described for an effective decision-support tool. Together, these themes specify where current practice falls short and what a tool would need to provide.

4.3.1 Theme 4: Current Foresight Practices and Gaps

Theme 4 examines the capabilities that boards express they currently lack or think could work better. Most interviewees recognized foresight as important to effective strategic oversight. However, the interviewees also portray it as a capability that is unevenly distributed across boards, conducted informally and individually rather than through any structured collective process.

The need for forward-looking

The demand for foresight was expressed most directly by the *Expert Advisory (I11)* who framed it as a response to a structural shift in the operating environment:

> *“The world is becoming… or really been growing in terms of uncertainty. The context that we live in [e.g.] it's AI, It's sustainability, it's different armed conflicts, and Tariffs. There's so much uncertainties. Doing business has become really much more complex than it used to be”* - Expert Advisory (I11)

Expert Advisory (I11) argument was that uncertainty at this level can only be addressed by a deliberate analytical practice: "the only way to deal with uncertainty is… tools to explore the future where you potentially have different scenarios different possibility, different idea of what might happen, and you start exploring them as you develop" (I11). The *Former CEO/Chairman, Pharma (I7)* described the same demand in evaluative terms, observing that boards he had served on had repeatedly been "surprised" by developments that could have been predicted with the right kind of imagination. *Former CFO, Tech (I5)* discussed foresight from previous experience as CFO, explaining that foresight must be seen as a mindset shift, which is required across the company.

> *“You need to move the angle to be looking ahead instead… it's not about hiring new foresight people. It's that the whole mindset of the company must be… what does it mean for the future? And the same goes for boards”* - Former CFO, Tech (I5).

Despite the recognized demand, several interviewees described foresight at board level as either largely absent or conducted on an individual basis without structured collective discussion.

Foresight activities such as scenario building and structured environmental scanning are described being done by the management rather than the board, even when boards engage with their inputs.

> *"We might not usually do that in board levels, but it can be also slowly growing into it. And usually, this is, viewed as operative and only done by management"* - Board Member, Fintech (I8)

Where individual board members do conduct forward-looking analysis, the practice is informal. Several interviewees described wanting to do their own research to get an individual opinion on the topic(s).

> *"...but at least try to go your get your own opinion. Otherwise, you feel like you're just rubber-stamping whatever material is coming out [from the management]"* - Former CFO, Tech (I5)

These individual practices have a compensating function similar to the supplementation behavior described in Theme 1, but they have an additional limitation that is specific to foresight: the resulting environmental awareness is not synthesized across the board. Board members arrive with different readings of the same environment, and these readings do not necessarily lead to a shared forward-looking picture during discussions. Foresight, on this account, exists at the level of individual board members but rarely as a structured collective capability.

The "who should be responsible" question

The interviewees were not in clear agreement on where the responsibility for foresight should sit, and the disagreement is itself a result. One position treats foresight as operationally located within management, with the board engaging with its outputs rather than producing its own analysis. A second position, treats foresight as something boards are responsible for catalyzing within management without doing themselves

> *"Boards are there to push and encourage management to come with [foresight]"* - Former CFO, Tech (I5).

A third position treats independent board-level foresight as a precondition for the board's challenge function, in which foresight serves the board as a tool to validate, challenge, or extend management's strategy. When asked which of these roles foresight served, I1 confirmed: 'All of them' (I1).

The three positions imply different design demands on the board's analytical infrastructure. If foresight is managements territory, the board's information needs are met by reviewing management outputs. If the board is responsible for catalyzing foresight, it needs analytical literacy to evaluate quality and push for accuracy. If the board needs independent foresight capacity, it needs access to forward-looking analysis that does not pass exclusively through management. The data does not resolve which of these positions is the better option, and the disagreement is presented here as a substantive finding rather than a difficulty to be smoothed over.

The limits of current scenario work

Several interviewees described scenario building as an activity they had participated in but argued that the work should be done by the management. This was partly related to scenario building being time consuming, and some argued out of the board's necessity. However, relying on management to create scenarios might be misleading, as one participant pointed out that these could be too optimistic.

> *"We were happy if we could create one scenario in the past. Very often, quite optimistic scenario was created. Maybe it didn't put up all the things which could go wrong, and the consequences of things didn't double up exactly as we planned and so on. And then every time something didn't happen as we had planned, we were surprised"* - Former CEO/Chairman, Pharma (I7)

The same interviewee was one of the few that had engaged in scenario building with board engagement. However, the tool was of quantitative characteristics and used in earlier decades and had some limitations. The output of the tool produced was a percentage of risk and percentage of success but did not address what the outcome would be if the risk were to happen. The gap *Former CEO/Chairman, Pharma (I7)* identified was in the methodology. There was an absence of narrative-level scenario work that traced consequences through the organization. *Former CEO/Chairman, Pharma (I7)* expressed an optimism about newer analytical tools, such as AI, being able to produce scenarios that are "more useful for management and boards by supporting navigation through pros and cons rather than only generating probabilities".

A final tension emerged regarding the appropriate trigger for foresight activities. This finding echoes the disagreement identified in Theme 2 but applied specifically to scenario work. While two interviewees argued that boards must adopt proactive, continuous foresight scanning in response to a dynamic environment, four others took a different stance. They expressed that tools like scenario building are too resource-intensive for routine meetings and should instead be triggered by actual, identified environmental signals. This disagreement raises the question of whether scenario work should be a continuous process or a reactive response.

4.3.2 Theme 5: Decision Support Requirements

Theme 5 describes the specific functional requirements that the interviewees mentioned were needed for an effective decision-support for boards. These functional requirements are here described as practitioner-stated requirements rather than observations with existing products. These descriptions explain what the interviewees would need a decision-support tool to do, and suggestion where the current foresight scanning falls short or could become more efficient.

Cross-domain synthesis tied to company context

The most voiced requirement was for an analytical tool that integrates signals across multiple strategic domains and is tailored to the specific company, that is able to produce analytical outputs rather than the raw data that is accessed by the board today.

> *"An AI agent that is aware that knows your company, knows the industry or sector you're in, including its technological development, its competitors' external shifts, geopolitical demographic"* - Former CFO, Tech (I5).

Correspondingly, the interviewees expressed the need that the tool must be adjusted to the type of business it operates within. For example, the relevant variables in pharmaceutical strategy, such as approval rates, pricing, and competition, are different from those in financial services or industrial manufacturing. Therefore, the primary requirement for cross-domain synthesis is a tool that is not a generic search engine, but one that is programmed with awareness of the company's specific industry context.

Layered format and pedagogical presentation

A second requirement discussed was how the output should be presented. Before the interviewees were given any description of a specific tool design, two of them expressed a layered format as a good approach based on their own experience.

> *"A kind of executive summary first... from this executive summary, there is more detailed, deeper and deeper detailed information about the different things"* - Former CEO, MedTech (I6).

Additionally, a related requirement was articulated by one of the participants, that however the format it is critical that the insights were to be presented in a pedagogical way.

> *"You can't or overestimate the importance of actually providing the insights or whatever is presented in a pedagogical way, pedagogy is critical"* - Fintech Chairman (I1).

When participants were presented with a description of how a layered tool might work, several interviewees further endorsed such format, while adding points they would appreciate. Some expressed interest of the idea of a dashboard *(I1, I4, I7),* to be able to see figures and click on it for more description. *Board Member, Fintech (I8)* did not like the idea of a dashboard based on own experience from other products, saying the user interface (UI) usually is confusing, and therefore an important requirement to be met for them to use a dashboard.

Identifying the Decision-Support Tools Use-Case

A third pattern was regarding when and how the tool would be used. Several interviewees pointed out that the tool's primary use-case is in pre-meeting preparation rather than as a tool used during board meetings.

> *"I would actually see the tool more in preparation rather than during the meeting. Because I don't want to be distracted by it"* - Board Member, Energy (I2).

> *"I think this tool would be useful before a strategy session. I think at home, at the kitchen table, each board member should kind of, well, be fed by whatever it is. And in whatever format it is. And then you do it collective So every board member comes into a joint session already warmed up"* - Fintech Chairman (I1).

With some interviewees adding the concern that a tool used during the meetings would consume time that is already scarce. One interviewee took the opposite position, arguing that a tool used during the meeting itself could add value in supporting or challenging the CEO's argumentation at the moment of decision, by providing cited sources for the recommendation or additional point of views. While other interviewees described the tool as being valuable before and during strategic days. These disagreements reflect different preferences and visions and do not necessarily point towards one option being better than another.

When discussing how the tool could be used, some interesting additional points emerged. One interviewee described how the tool could act as a "generative challenger" where the tool could help the board stress-test the managements strategy. Same interviewee noted that some sectors are already "developing strategic scenarios with the help of AI to discuss" *(I1)*. Another interviewee emphasized that the value comes from interaction with the tool, rather than from one produced output.

> *"As usual with AI... it's a dialogue you have. And it's the interaction that makes it great"* - Former CFO, Tech (I5).

This suggests that the value of such a tool lies in interactivity rather than static output. Whether used for individual preparation or collaborative discussion, the interaction and dialog with the AI may allow boards to independently prepare according to their own preferences.

4.4 The Adoption Layer

The three themes in this layer together describe the conditions board members identified as necessary for an external analytical tool to become part of their board work. Unlike the themes in the previous layers, which describe what currently exists, these themes describe what needs to be true for adoption to take place. Theme 6 examines the trust conditions under which board members would rely on AI-generated analysis. Theme 7 sets out the parts of board work that interviewees argued cannot be handed over to a tool, no matter how capable it becomes. Finally, Theme 8 turns to the organizational conditions, including AI literacy, ownership, and behavior change, that determine whether adoption takes place or not. Together, these themes specify what needs to be true for a tool of this kind to be used rather than ignored at the board level.

4.4.1 Theme 6: Trust Architecture for Board-Level Tools

Theme 6 examines circumstances in which the interviewees would consider an external analytical tool trustworthy enough for board-level use. Most participants described trust as something that develops gradually through accumulated performance, supported by structural features that allow board members to verify outputs against original sources and allow for independent conclusion taking on what they read.

Accuracy as non-negotiable condition

One of the trust barriers that was identified consistently across the interviews was the risk of hallucination. One of the interviewees described a situation where their board tried to implement a "board-GPT" assistant, but with undesired results. The board did not use it, which the interviewee believed was a result of the board members not trusting the tool; members feared potential hallucinations and doubted the tool's ability to produce strictly fact-based outputs. This scenario where a tool that is available but actively ignored illustrates a trust failure at the architectural level, likely stemming from failing to meet the technical expectations.

It is understandable that boards might be more cautious to an AI assisted tool, as the cost of acting on incorrect output in board context can be significant. However, several interviewees added that each individual board member needs to apply their own competence to identify whether it is hallucination or real.

Trust is accumulated over time

Several interviewees described trust as something that accumulates over time through repeated accurate outputs rather than something the tool can establish at the point when it is introduced.

> *"I would certainly read the output of the AI. And after a year, if I find the tool really useful, I would trust it more. I would probably tune down my own reading as preparation work [as a result]. And, of course, that's the purpose, but you need to build that experience and confidence in the system before you go there"* - Former Head of R&D, Pharma/Biotech

> *"Maybe you trust some brands more than others. If they are reputable, well companies that you worked with before... you tend to maybe accept it because they have provided results before"* - Former CFO, Tech (I5).

The shared view is that trust towards AI as a decision-support tool comes gradually, similarly to anything that is new. Trust can be increased through reliability and accuracy and decreased through observed failure. The provided insight implies that a tool must be usable even under low-trust conditions, since high-trust will only be reached, if at all, after a longer period of cross-checking by users who did not initially rely on it.

Verification and Calibration of AI

To allow for trust to be accumulated over time, source transparency was identified as a baseline requirement. Several interviewees expressed the importance of being able to trace back to original source if something felt wrong and confirm accuracy independently of AI.

> *"You need to be sure that it's real facts and not just made-up facts... if you have references to the sources, you avoid that type of suspicion"* - Former CEO/Chairman, Pharma (I7).

When the participants were presented with the concept of a confidence rating score, a majority of the interviewees reacted positively. *Former Head of R&D, Pharma/Biotech (I4)* added similar ratings had been implemented by the management team at one of the organizations, where the rated score was based on number of sources.

> *"They have also ratings on the level of certainty about certain information, so to score really high you needed to have three different sources [...] I think it is helpful basically, in particular if you indicate that this is uncertain information"*- Former Head of R&D, Pharma/Biotech (I4)

Interestingly, when asked directly, a majority of the interviewees did not express significant concern regarding trusting a verified AI tool. Several of the participants pointed out that human employees and management teams also make mistakes, carry biases, and "hallucinate" in their own ways.

> *"Many tools nowadays hallucinate, but everyone hallucinates, even the humans"* - Board Member, Fintech (I8)

Another aspect pointed out by one interviewee was that absolute precision is not always required for such a tool. They exemplified this by noting that if the input were to get a countries' population, but the output failed to give the exact number, the error would most likely not make a significant difference for strategic purposes.

When discussing the security and governance of such a tool, most participants expressed concerns about inputting internal data, as this must be done with extreme caution. However, they recognized that this concern is based on the current state of AI and might become irrelevant in the near future as technology continuously improves.

The combined insights from the interviewees indicate that; while transparency and signaling for uncertain output are desired requirements, the tool output does not always need to produce flawless outputs. Reason being that board members are already familiar with cross-checking the management, who also make mistakes, meaning that the same practice would be applied to AI.

4.4.2 Theme 7: Human Judgment as Non-Negotiable

Theme 7 examines the specific board functions that interviewees believe cannot be outsourced to AI, no matter how capable the technology gets. First, they argued that the tool serves only as an advisor, leaving the final choice to the board. Second, all legal and reputational risks are inherently human and cannot be outsourced. Finally, evaluating context goes beyond data sheets; it is about applying accumulated experience and relying on what one interviewee called a 'gut feeling'.

The tool as support, not actor

When discussing the tool's primary function, a majority of the interviewees pointed out a clear boundary between what the AI can do and what the board must do. While the tool can handle extensive analytical work, the board's ultimate decision-making responsibilities cannot be outsourced.

> *"The role of the board is to make judgment calls [...] I don't exactly foresee AI doing that... to replace the board members at all, but I see AI being a tremendous lead forward in supporting the board in that aspect"* - Board Member, Energy (I2).

This exemplifies a human-in-the-loop approach: the AI provides the data, but the board retains the responsibility for turning that data into strategic judgments. This separation is naturally tied to legal accountability. Interviewees pointed out that one cannot brush off a bad outcome by saying "it was just a computer" *(i.e. I7)*.

> *"If I follow [some] advice from an AI, it's not the AI's responsibility, it's my head that will roll"* - Head of Innovation, Digital Health (I9)

Consequently, integrating an AI tool at board-level does not reduce the need for individual members' expertise. Instead, using such a tool might test the board's own competence, challenging them to actively evaluate the AI's output rather than passively accepting its conclusions. It requires the board members to maintain a critical mindset and not blindly sacrifice their own judgment when presented with the AI's answers.

> *"You need to be critical and to understand it's only a tool... not sacrifice their own competence when the AI gives answers. And this ability and the competence of the board is what's tested, even when we use an AI tool to support their decision"* - Former CEO, MedTech (I6).

The "gut feeling" beyond the tool's reach

On the topic of AI as a support-tool for strategic decisions, two interviewees pointed out that the process involves more than what external and internal data captures. *Head of Innovation, Digital Health (I9)* expressed that while AI is "very logical, very rational", human decisions regarding strategy involve non-translatable variables like "dreams, hope, wishes, [and] thoughts". Consequently, an AI tool can only provide "a piece of the puzzle" as informal sentiments and undocumented organizational signals remains inaccessible to it. *Head of Innovation, Digital Health (I9)* further linked this to the magnitude of choice, while routine decisions rely heavily on data; bigger strategic choices require an intuitive instinct that AI cannot provide.

> *"The bigger the decision, the more gut feeling... The smaller the decision, the more fact-based it is"* - Head of Innovation, Digital Health (I9)

Similarly, *Board Member, Energy (I2)* described their own board preparation as "experience-based primarily." They explained that their background guided their thinking process when evaluating a proposal, whereas a tool like AI is merely used to "quick search for certain topics" when a subject feels less familiar.

Taken these insights together, the tool is best positioned as a supplement for information gathering. It suggests that, at least for some board members, complex strategic choices will always require a level of human "gut feeling" that a machine cannot provide.

4.4.3 Theme 8: Conditions for Adoption

Theme 8 examines the conditions that are needed for a board to truly consider adopting and using an external AI based analysis tool. The results from the interviews suggest that even a tool meeting the requirements mentioned in Theme 5 and the trust conditions discussed under Theme 6 will not be adopted unless three further conditions are met: a baseline level of AI literacy across the board, ownership over how the tool is managed, and an effort to shift board routines to integrate such AI tool.

Literacy as the baseline condition

Several interviewees identified AI literacy as a precondition for any tool to be used effectively by board members, both individually and collectively. However, a less optimistic view of readiness was observed, with some interviewees pointing out that most boards "are not on top of it yet" and that current usage and competence was barely "scratching the surface" (i.e., I5).

> *"Step one would actually be for every board member's to actually kind of get to know AI as a tool. Otherwise, they would be super suspicious to it [...] and would challenge it [the output] immediately. Especially from more old school people. So, I think you need to build that awareness, otherwise, it would be hopeless"* - Fintech Chairman (I1).

A related question was whether the literacy requirement was better resolved by adding an AI specialist to the board or by merely increasing the AI competence of existing members. There was a clear division in this aspect, where some of the interviewees stated that the literacy did not have to be on a specialty level, and that including a specialist may disrupt the knowledge composition of the board.

> *"The question comes up, maybe we should have an AI expert... But then after some discussion, the answer is almost every time 'we can bring in an expert to the board meeting' and so on. But to have somebody on the board who is, like, super deep in one topic, but maybe doesn't know about everything else related to governance in the board, that doesn't really help"* - Former CFO, Tech (I5).

It was however necessary for board members to have a baseline understanding of the operational aspect of AI to better grasp the strategic opportunities it may provide. In contrast to this, some of the interviewees argued that, in addition to baseline literacy, future board recruitment should actively address this gap as well. Framing that part of the literacy demand is inevitable linked to bringing competence to the board.

> *"When we recruit new members to the board, that might change the board structure itself because newer competencies like AI and environment regulatory issues have an impact on development in skills and better adoption of new features"* - Former CEO, MedTech (I6).

Consequently, the requirement for AI literacy may need to be shaped both with immediate training and long-term strategic recruitment. While current members need a baseline understanding to adopt the tool, the board's may need to consider expanding their knowledge composition as AI competence might evolve into a core requirement in any context

Internal ownership and accountability

A second adoption condition addresses who within the organization should be accountable for the tool. None of the interviewees raised this topic spontaneously, but when asked about it directly, two accountability questions emerged. The first concerns who should operate and monitor the tool itself. Some of the interviewees identified roles like chief digital officer, CIO, or other relevant business unit head as the appropriate owner.

> *"It sounds like a good idea, to actually have somebody with accountability of managing a tool like this... a chief information officer to monitor it. That might increase the trust"* - Former CFO, Tech (I5).

While pointing out that such practices already exist within the organization in regard to other tools, and therefore same practices should be applied to any other tool.

> *"We always try to have accountability for all tools that are used in the organization. The one who's generally responsible for all these services is CIO. And if it's a specific niche business tool, it could be someone in the business area product development"* - Board Member, Fintech (I8).

The second concerns were in regards of who, if anyone, should review the tool's output before it reaches the board. Several interviewees converged to the CEO as an appropriate reviewer, each with a different reason. One aspect was in regards of protecting the board meeting time *(I3),* another aspect was in regards of the tool gaining trust *(I5, I6),* while a third aspect being ensuring quality and legal accountability of the output *(I6, I7).*

In contrast to this, one interviewee argued that a formal review process is unnecessary. They pointed out that boards have never had oversight over how, example given, management produce their material and analysis; employees use Google, research papers, and use networking as ways of gathering information. The board's job is to evaluate the final material and not to oversee the process behind it. This divergent perspective does not necessarily reject the need for accountability but rather views it from a different angle. Instead of requiring a new AI-specific review procedure, it suggests that AI-generated data should merely be evaluated with the same critical lens the board already applies to human-generated outputs.

The risk of adoption without behavior change

A third concern was in regard to the risk that might follow if such a tool is adopted, but the board's actual work does not change. Interviewees highlighted two main aspects of this risk. The first is that an AI tool must be embedded into the organization to change its core operation. Otherwise, boards risk adopting it on a superficial level.

> *"The big risk is that it doesn't change the way you're doing your core business... using an AI tool that will not be kind of implemented or embedded or embraced by the organization. It's seen upon something [e.g.] - 'So Claude made that... Okay. Let's do it our way' "* - Fintech Chairman (I1).

The second aspect relates to failing to take full advantage of the tool. One interviewee pointed out that since AI has the potential to monitor data in real-time, the traditional infrequent strategy meetings could essentially limit the technology's potential. These aspects point towards the same underlying problem: adoption of new technology fails if the board refuses to update the traditional way of working.

A possible reason why these habits are hard to break relates to the individual board members' professional identities. Several interviewees acknowledged that most board members are seniors but did not identify any generational barriers as a risk for adoption. They argued that it would rather be a matter of "personal interest" and openness to change.

> "*Many board members are used to having all the answers. They are so experienced, they've been through similar situations previously, they are fact-oriented, they are creative... which is of course extremely valuable. But that also makes it sometimes hard to adopt to new situations"* - Head of Innovation, Digital Health (I9).

Nevertheless, changes have happened before, and initial resistance is inevitable according to some of the interviewees. One of them added that if a tool proves valuable, those who resist will eventually

be forced to adapt. However, overcoming this friction cannot be expected to happen automatically and it requires active change management with dedicated time for learning, and a structured approach

> *"You need training. So, when you introduce the tool, you need to offer a e.g. two-hour training course… if you just throw it on people that don't know how to use it, it's kind of a hurdle. They just postpone it to next time"* - Former Head of R&D, Pharma/Biotech (I4).

Taken these together, the findings illustrate a paradox: the experience that makes board members valuable might be the very same reason that causes resistance. Successful adoption might be less about teaching the technical features of the tool, and more about guiding experienced board members to let go of old habits and embrace new approaches.

5 Results – The Design Framework

This chapter presents the main result of the thesis: a design framework including a set of design principles for the AI foresight tool, a decision-support tool intended to support board members in their strategic foresight discussions and environmental scanning. The framework was developed through an iterative synthesis of the theoretical literature reviewed and the empirical findings from ten semi-structured interviews and one expert advisory session in Chapter 4. Thematic analysis of the interviews uncovered eight themes: information architecture, the episodic governance trap, strategic days as an integration point, current foresight practices and gaps, decision support requirements, trust architecture, human judgment, and conditions for adoption. The framework is structured around four design dimensions, derived from the theoretical background described in Section 2.8, and extended to 21 design principles.

The framework is a design-oriented knowledge contribution; it does not predict outcomes but specifies what a well-designed tool should do and why. Its claims are grounded in a combination of theoretical reasoning and practitioner validation from board directors across different sectors, including fintech, energy, technology, pharma/biotech, medtech, and digital health.

5.1 Design Framework Overview and Structure

The management and governance literature reviewed in the theoretical background section identifies two categories of challenges that a board-level AI foresight tool will or must address. The first is answering research questions one (RQ1), concerns the *information and foresight environment*: what gaps do boards face in information and foresight practice, and what must a tool do to address them? The second, referring to RQ2, concerns the *adoption and governance environment*: under what conditions will boards trust and use such a tool, and what design features ensure that its use remains legally and deliberatively sound?

5.1.1 From Layered Empirical Findings to Design Dimensions

The thematic analysis of the interview data in Chapter 4 were grouped into a *context layer*, a *capability layer*, and an *adoption layer* that together describe the current state of board information work and the conditions under which AI support would be adopted.

The design framework presented in this chapter is organized with evident requirement recognized from the empirical findings (Chapter 4), but the framework is structured differently. It is structured around four design dimensions, each containing a set of design principles, reflecting the design decisions that must be made when specifying an AI tool for board-level foresight. Chapter 4 describes the problem experienced by the board members, while Chapter 5 specifies the requirements the tool must fulfill. This shift in organizing is due to how they serve different analytical purposes: themes are grouped in descriptive categories that describe what interviewees experienced, while dimensions are grouped in categories that describes what the tool must do.

Table 5.1 below maps each empirical theme to the design dimensions (D) and specific design principles (DP) it informs. A theme can inform multiple dimensions, and a single dimension can draw multiple themes; the mapping is therefore many-to-many rather than a one-to-one correspondence. The purpose of the table is to make the interpretive link between what was found (Chapter 4) and what was designed (Chapter 5) traceable. Each principle developed in the following sections cites the specific empirical findings and theoretical literature that ground it.

A full design principle traceability matrix, listing every principle with its empirical and theoretical grounding, is provided in Appendix E.

Table 5.1: Mapping between themes and design dimensions/principles

Theme	Dimension, D (Design Principle, DP)
Theme 1: Information Architecture Problem	D1 (DP1); D2 (DP7)
Theme 2: Episodic Governance Trap	D2 (DP10)
Theme 3: Strategic Days as Integration Point	D1 (DP4); D2 (DP8); D4 (DP18)
Theme 4: Current Foresight Practices and Gaps	D2 (DP8, DP9, DP11)
Theme 5: Decision Support Requirements	D1 (DP1, DP2, DP5, DP6); D2 (DP9)
Theme 6: Trust Architecture	D3 (DP12, DP13, DP14, DP15)
Theme 7: Boundaries of Human Judgement	D4 (DP17, DP18, DP21)
Theme 8: Conditions for Adoption	D3 (DP15, DP16); D4 (DP19, DP20)

5.1.2 What is a Design Principle?

A design principle, as used in this thesis, is about how a system should be built, grounded in both theoretical reasoning and empirical evidence (Gregor and Hevner, 2013). It takes the form: the tool should do X, because Y (theoretical justification, where relevant) and because Z (empirical finding quoted and cited). Each principle in this framework satisfies three criteria. First, it is grounded: it can be traced to at least one theoretical source/ at least one empirical finding from the interviews, identified in the principle statement. Second, it is actionable: a developer reading it can make a concrete implementation decision based on it. Third, it is bounded: it specifies not just what the tool should do but what it should not do, where relevant, to avoid the governance failures the literature and interview data identify.

5.2 The Four Design Dimensions

The framework is organized around four design dimensions, each grouping a set of design principles that address a distinct aspect of what a board-level foresight tool must do. Together they span the two challenge categories introduced in Section 5.1: the information and foresight environment (RQ1) and the adoption and governance environment (RQ2). The four dimensions, cognitive augmentation, foresight function, trust and adoption, and governance and accountability, are summarized in Figure 5.1 and detailed below

Dimension 1 - Cognitive Augmentation Requirements: The literature on bounded rationality and information overload discusses the same pattern, when boards face more information than they can process, they satisfice rather than analyze, settling on what is in front of them rather than searching further (Simon, 1955; Ward, 2023). This dimension specifies how the tool must present information so that it extends board capacity rather than burdens it.

Dimension 2 - Foresight Function Requirements: Boards routinely miss peripheral signals from adjacent industries and early indications of disruption because information has already passed through management's interpretive filter before reaching the boardroom (Daft and Weick, 1984; Choo, 1994; Robinson and Simmons, 2018). This dimension specifies what the tool must scan, surface, and simulate to deliver foresight intelligence that reaches the board in a form directors can use.

Dimension 3 - Trust and Adoption Requirements: Adoption depends not on the technical quality of the tool alone but on how directors perceive its usefulness, how their peers signal its value, and how familiarity with the technology builds over time (Davis, 1989; Venkatesh et al., 2003; Marocco, Barbieri and Talamo, 2024). This dimension specifies the verification mechanisms,

deployment architecture, and change-management conditions required for the tool to be adopted and trusted in practice.

Dimension 4 - Governance and Accountability Requirements: Board decisions must remain attributable to human directors, AI use is constrained by fiduciary duty, and unchallenged outputs risk epistemic capture (Ben-Shahar *et al.*, 2025).

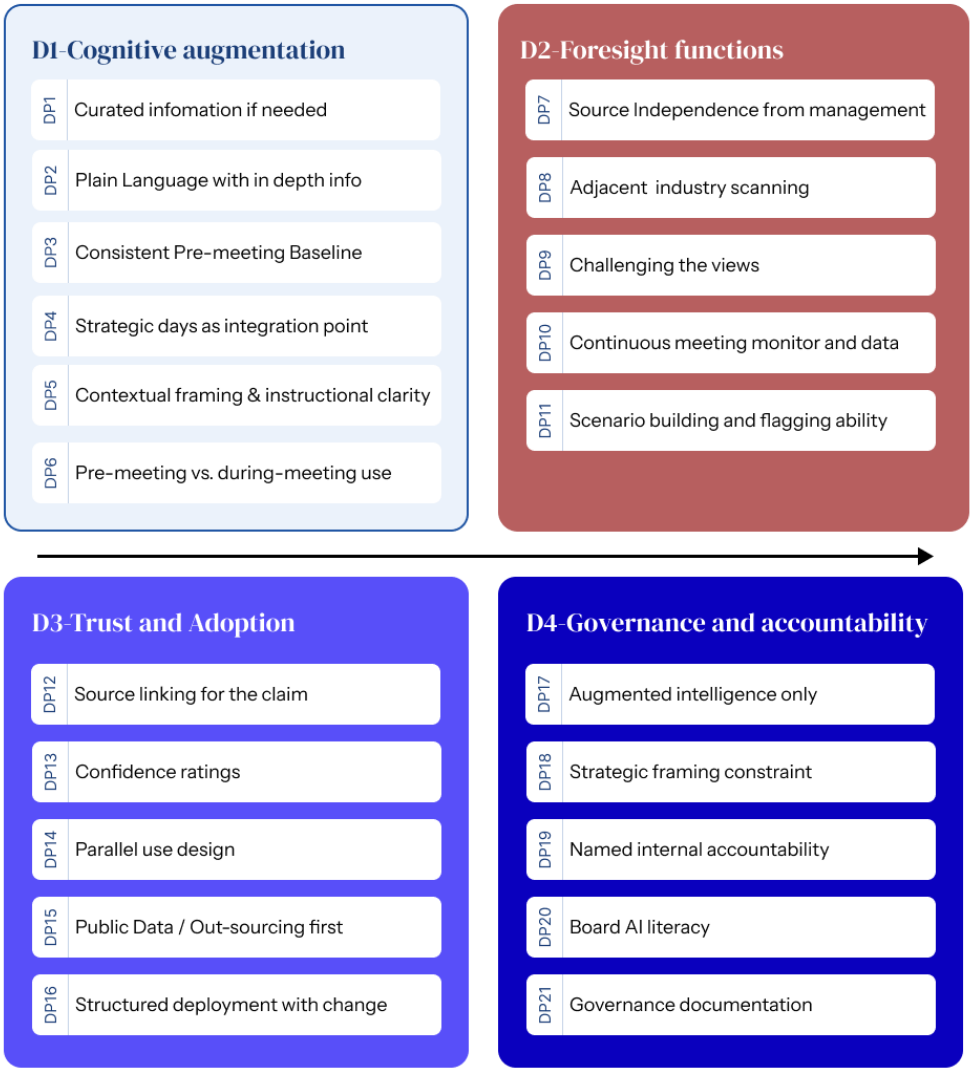


Figure 5.1: **Four-dimension design framework for an AI-based foresight tool for corporate boards, with 21 design principles derived from theoretical literature and empirical interviews.**

5.2.1 Dimension 1: Cognitive Augmentation Requirements

Dimension 1 addresses the cognitive conditions under which board members process strategic information. The theoretical grounding for this dimension draws on Simon's (1955) account of bounded rationality, which holds that decision-makers satisfice when information exceeds cognitive capacity, and on Ward's (2023) extension that information abundance can produce inattention rather than better decisions. The six principles in this dimension specify how the tools output should be structured to support rather than burden board cognition.

Curation over comprehensiveness (DP1)

The tool provides three to five selected signals for each cycle of meetings, arranged according to their strategic importance and evidential strength. The number of signals is a framework decision in response to the volume problem, not an empirical finding. As seen from Theme 1, board members indicated that comprehensiveness results in paralysis instead of clarity; what is needed is precision and quality, not volume.

This view supports both Simon's (1955) theory on how decision makers stop searching when an item crosses a certain threshold and another study showing how an excess number of options affects the quality of decision-making (Ward, 2023). The tool selects and sorts; it does not combine and dump. There is a limit on the output size per cycle regardless of the actual number of identified signals. Unrestricted output is a bad design flaw, not an advantage.

Plain language with layered depth (DP2)

This tool ensures that the executive summary is comprehensible by all board members, with an additional level of analysis for the curious ones if needed. According to Theme 5, the correct presentation method in terms of layers was stated independently by two interviewees without any tool being introduced prior to their interview, while another interviewee reiterated the importance of pedagogical clarity.

This is consistent with the literature which says that explainability should be aimed at the target audience, i.e., verification, rather than technical explainability in the case of board members (Barredo Arrieta et al., 2020). Hence, the tool will provide a double-layered output the first one is the board-level summary, and the second is the drill-down level, both delivered at once in the same document (Rieger et al., 2025).

Consistent pre-meeting baseline for all board members (DP3)

The tool provides the same external intelligence report to all the board members prior to each meeting regardless of their personal tendencies for conducting research. As discussed in Theme 1, personal tendencies for supplementing information differed among board members, where some conduct extensive personal research before each meeting while others do not. This may result in unequal analytical backgrounds for the same board members.

Forbes and Milliken (1999) argue that a good board discussion necessitates equal inputs of information. Any form of inequality in information input will hamper the knowledge-utilization norm of collaborative decision making (Ben-Shahar *et al.*, 2025). The tool's response is to distribute the same external intelligence report to all board members at once.

Strategic days as the primary integration point (DP4)

The tool creates a macro environment briefing customized for the strategic funnel's external orientation, two weeks ahead of annual or biannual strategy sessions. As seen in Theme 3, strategic days constitute the major point of integration in the governance year, while the external orientation of the strategic funnel is exactly the part where management inputs are likely to be sector-specific.

Foresight intelligence routinely fails to reach the boardroom at the synthesis stage, where environmental signals must be translated into something a board can deliberate on (Choo, 1994; Rohrbeck and Gemünden, 2011; Li *et al.*, 2018). The tool therefore creates an exclusive strategic-day briefing with wider coverage than ordinary pre-meeting briefings, including macro environment, adjacent sectors, and regulation. It does not wait for the strategic day to commence; the external baseline must be ready before the directors walk into the room.

Contextual framing and instructional clarity (DP5)

The tool classifies every signal into a thematic context with an explanation of the strategic significance it contains in clear language before it reaches the board. As Theme 5 indicates, respondents uniformly stated that raw data was insufficient. Rather, what was needed was some form of synthesis that resulted in meaning, and not information that demanded synthesis by the user. Where organizations react to what is interpreted rather than the environment itself.

Daft and Weick (1984) frame the same point at the organizational level: organizations respond not to the environment itself but to their interpretation of it, and interpretation is what creates the organizational perspective (Daft and Weick, 1984; Choo, 1994). The design implication is that the tool never presents raw data without the context in which it should be read. Every output answers the same underlying question: what does this signal mean for this firm, in this industry, at this particular time. The tool does not deliver data and call it analysis; leaving the board to interpret raw scan output would simply reproduce the synthesis problem the foresight literature already identifies. Pedagogical framing transforms a received signal into an opportunity for the board.

Pre-meeting vs. during-meeting use (DP6)

The principle of pre-meeting preparation opposes an intuition that many boards initially adopt regarding the possibility of using the tool during meetings. The empirical data found in Theme 5 is unambiguous yet opposed. While one of the interviewees found the tool useful for supporting the decision-making process during meetings, the majority of data suggests that the tool should be used only for preparing before the meetings; its use during the meetings should be limited to the CEO-prepared analysis presentations.

The determining factor in this case is the time factor, as when the tool starts to suggest new threads for analysis, it becomes a competitor of the board's agenda in terms of spending time, whereas it is not the case with pre-meeting preparation tools. This constraint has a theoretical basis in two directions: Forbes and Milliken's (1999) finding that effort norms the quality of director preparation directly mediate decision quality, meaning pre-meeting use is the stage at which the tool can most productively influence deliberation; and Simon's (1955) account of satisficing under time pressure, which predicts that introducing new analytical threads during a time-constrained meeting will reduce rather than improve the quality of the decisions reached.

5.2.2 Dimension 2: Foresight Function Requirements

Dimension 2 addresses what the tool must scan, surface, and simulate to fill the foresight gaps identified in Chapter 4. The theoretical grounding draws on Daft and Weick's (1984) account of strategic environments and the strategic blindness produced by narrow scanning, and on Forbes and Milliken's (1999) framework of cognitive disagreement as a condition of effective board decision-making. The five principles in this dimension specify what the tool must look for, how broadly it scans, and what kinds of alternative readings it must surface.

Source/epistemic independence from the management channel (DP7)

The tool scans external source without referring to management editorial perspective or the content of any management produced board documents. As indicated by Theme 1, the ten participants uniformly explained that pre-meeting material passes only through the CEO or management prior going to the board, resulting in a case of information asymmetry in which the cognitive context set by management shapes what the board sees. Large firms in particular tend to fall into conditioned viewing, with reporting systems built to address known issues and therefore producing predictable blind spots around what those systems were not designed to surface (Daft and Weick, 1984; Engstam *et al.*, 2024). McNulty and Pettigrew (1999) link this directly to board influence: directors exercise more meaningful strategic influence when they have independent analytical access to the environment rather than depending on management's framing of it.

Thus, the tool has access to the primary source material directly, without summarizing or synthesizing management documents. The tool acts as a secondary channel, not subject to management at all. The tool does not begin from management's strategic perspective, since to do so would perpetuate the tunnel vision that Theme 1 suggests is the biggest structural constraint on board information.

Weak signal detection and adjacent-industry scanning (DP8)

The scanning tool does not confine itself to the immediate competitive landscape but extends to include other industries, new frameworks for regulation, and indications coming from non-obvious competitors. As is evident from the Themes 3 and 4 analyses, both boards and smaller companies stop at the sector-level only, as illustrated by Theme 3 in which it was found out that the interviews mentioned active importation of external perspectives to compensate for this limitation.

Conditioned perception will lead to blind spots regarding any novel disruptions; similarly, the internal reports tend to underestimate indications coming from adjacent domains, and the weak detection of peripheral signals is a recognized consequence of scanning bounded by the current strategic frame (Daft and Weick, 1984; Robinson and Simmons, 2018; Grove, Clouse and Xu, 2023).

The scan will thus incorporate at least one adjacent industry indicator and one signal from the macro-environment during each round, independent of their presence in the board's existing strategic frame. In other words, the scan does not limit itself to what the current competitive set of the organization comprises. An equivalent scan would simply reinforce existing views.

Challenging perspectives and flagging issues (DP9)

The tool actively searches out and brings to light any evidence that may go against or challenge the assumptions implicit in the management's strategic plan. As indicated in Theme 4 and 5, while analyzing the scenario elements in relation to the reactions, the most frequently liked scenario for all the ten participants was the part that challenged the strategic assumptions made by management.

According to Forbes and Milliken (1999), constructive cognitive conflicts improve decision making by examining underlying assumptions. The opponent role perspective challenging and stress-testing existing strategic commitments is the foresight function most directly connected to

board challenge capacity, and the one least likely to be activated when foresight is managed exclusively within management (Rohrbeck and Gemünden, 2011). Ben-Shahar et al. (2025) extend this argument to AI specifically, identifying algorithmic groupthink as the failure mode that emerges when AI outputs go unchallenged, and pointing to a Devil's Advocate AI design that actively surfaces alternative readings rather than confirming a primary one.

As such, the tool makes provision for the contradiction element by presenting an opposing view of the current strategy as analyzed by the CEO using AI. The tool does not confirm the validity of management's analysis. If it did, it would be contradicting its primary purpose.

Continuous monitoring with configurable schedule (DP10)

The tool scans external signals constantly between meetings and provides structured briefing schedules based on a cadence that is tailored to fit the industry and governance cycle of the board. In Theme 2, the episodic cycle of governance does not correspond with the speed of disruption; the *Fintech Chairman (I1)* saw strategic plans as outdated from the second they were approved, and the energy industry board member described the dilemma of preferring stability but needing adaptability. The idea that foresight is only effective when it works as a dynamic capability constantly in use rather than an annual event (Haarhaus and Liening, 2020). At board level specifically, this means treating environmental sensing as an ongoing capability rather than a periodic activity bound to the strategy day or the annual review (Engstam *et al.*, 2024).

The tool operates on a constant basis to provide briefings at all times, even between meetings. Cadence is determined by the board prior to implementation and can be tailored depending on the industry; fast-paced boards might receive briefings once a week, while slower boards might receive briefings once a month. The tool cannot only work on a pre-meeting trigger. An application that only operates 72 hours before the meeting and sleeps afterward will replicate the episodic pitfall found in Theme 2.

Scenario building and consequence mapping (DP11)

It creates additional scenarios, including pessimistic scenarios, and tracks how failure in each of those scenarios will cascade into organizational consequences for other areas of the company. It is consistent with the Theme 4, where historical approaches to scenario creation were optimistic in nature and methodologically flawed, since the approach previously used provided probability estimates without creating a map of future events should they come to fruition within organizations.

The consequence of blindness being the primary failure of traditional scenarios and the need for exponential thinking in relation to non-linear futures (Marinković *et al.*, 2022; Grove, Clouse and Xu, 2023).

For each strategic issue, the tool creates at least three scenarios: a baseline, an optimistic one, and a pessimistic scenario, including a consequence map of second-order effects across two or more organizational processes. However, the tool does not only create optimistic or baseline scenarios. Creating only optimistic scenarios through a tool will simply reinforce the structural bias toward optimism outlined in the fourth theme as one of the core foresight failures.

5.2.3 Dimension 3: Trust and Adoption Requirements

Dimension 3 addresses the conditions under which boards would trust and ultimately adopt the tool. The theoretical grounding draws on the technology acceptance literature (Davis, 1989; Venkatesh *et al.*, 2003), Cooper et al.'s (2025) analysis of trust expectations in AI systems, and Marocco et al.'s (2024) work on AI adoption in organizational contexts. The five principles in this dimension specify the verification mechanisms, deployment architecture, and change management protocol required for the tool to be adopted in practice.

Source linking for every claim (DP12)

All items on the AI briefing are connected through a clickable link to the source document. As demonstrated in Theme 6, source transparency was the one criterion consistently highlighted as a necessary element by all ten participants.

It is consistent with Barredo Arrieta et al. (2020), who argue that, when it comes to the needs of managers and board members, explainability entails verifiability, i.e., the ability to verify whether a claim made is sensible and valid without needing any knowledge about the AI process. In turn, the source transparency provides the means through which directors can independently evaluate information generated by AI (Zeiser, 2024).

Thus, all claims, signals, and data points in every output have links to the source. No assertion is generated by the tool that cannot be sourced. An unverifiable assertion is an inherent risk.

Confidence ratings (DP13)

Every piece of information in the briefing is given a three-level confidence indicator, namely High, Medium, or Low. For Low-rated pieces, the reason behind the low confidence level is explained in plain language. It can be seen from Theme 6 that most interviewees reacted positively towards the confidence indicators; one of them said that he had a similar mechanism being used by a management group in which high confidence meant a need for three different sources.

The AI-induced trust can be described by an expectation-updating theory where inflated confidence ratings cause a higher drop in trust than accurate confidence. Furthermore, Rieger et al. (2025) state that explainability should determine action and not mere confidence levels, which means that ratings should help the director determine scrutiny level, not confidence.

The tool rates every single piece of information and avoids hiding low-confidence ratings. Directional usefulness and not factual precision should be the standard of accuracy for the board members during onboarding. The tool does not inflate its confidence ratings to appear authoritative. Inflated ratings are a trust killer.

Parallel-use design (DP14)

The tool is introduced as an addition to current board preparation activities rather than as a substitute, and this trust-building process lasts at least six months. It was found that in Theme 6 participants had a transition cycle whereby initially they would verify AI's outcomes using their own literature search; then, a year later when they had trusted AI's results, they would start reducing their reading time.

Fahnenstich, Rieger and Roesler (2024) identify the underlying reason this design is necessary: directors may state in principle that they trust an AI tool while still reverting to traditional preparation when an actual meeting approaches, since trust attitude and trust behavior diverge in practice. Familiarity built through repeated exposure under verifiable conditions is what converts attitude into behavior over time (Marocco, Barbieri and Talamo, 2024). Parallel use creates exactly that condition: board members combine the tool's output with their own sources, and the tool must be designed to work well under that cross-checking rather than under exclusive reliance.

Public data search by default (DP15)

The tool is restricted to using only public sources until an explicit board decision mandates inclusion of internal sources, supported by proper technical measures. As seen in Theme 6 and 8, some of the participants expressed a concern in regard to using internal data sources; one participant even equated data strategy to corporate strategy, which implies that accessing internal data is a governance issue rather than a technical one.

The risk associated with data privacy, as a legal consequence of using AI for board purposes, in terms of uploading the confidential data to the third-party system (Papagiannidis, Mikalef and Conboy, 2025). The proposed mitigation in the governance literature is to limit what the tool is able to process at all, treating the data-access boundary as itself a governance instrument (Chriki *et al.*, 2025).

The tool therefore needs an explicit board resolution before using internal sources. Default settings include only public sources, providing the boards the chance to familiarize themselves with the utility of the tool before dealing with the difficult governance aspect of internal sources.

Structured deployment with change management protocol (DP16)

The implementation of the tool follows a defined process that involves onboarding training, integration into the calendar, and a defined pilot phase, instead of being deployed simply as an IT initiative. According to Theme 8, participants recognized the potential problem of adopting without behavioral change as the main cause of deployment failures; the use of an absent tool can lead to the abandonment of the tool.

Usefulness is considered to be the main driver for adoption, which is possible only when usefulness becomes experienced by frequent interactions with the tool (Davis, 1989; Venkatesh *et al.*, 2003).

Therefore, the tool will be included into the agenda from the very beginning as an important component of the existing processes within meetings, such as pre-meeting briefings, quarterly scans, and strategy day preparation. A voluntary pilot phase without integration into the calendar cannot succeed even during its first phase.

5.2.4 Dimension 4: Governance and Accountability Requirements

Dimension 4 addresses the conditions under which the tool can be deployed without undermining the legal and deliberative legitimacy of board decision-making. The theoretical grounding draws on Mertens (2023) and Chriki et al. (2025) on the fiduciary constraints that govern AI use in corporate decision-making, and Ahdadou et al. (2025) on the augmented-intelligence stage at which AI may legitimately support board work, and on Ben-Shahar et al. (2025) on the risk of epistemic capture when AI outputs go unchallenged. The five principles in this dimension specify how the tool must be designed and governed to preserve the conditions that make board governance legitimate.

Augmented intelligence stage only (DP17)

The tool delivers analytics inputs that will enable director decision-making; it offers no recommendations, no evaluations, no conclusion that would indicate that AI had already formulated a strategic stance. As indicated by Theme 7, board members made a distinction between what AI could do and what was expected from the board, and some of them said in their interviews that the individual on the board would remain accountable for any results delivered by tools.

Augmented intelligence is the only stage of AI involvement compatible with the fiduciary basis of board work. The model is defined as one in which AI provides analysis but does not make decisions, with accountability remaining with the directors (Ahdadou, Aajly and Tahrouch, 2024; Agnese, Arduino and Di Prisco, 2025; Ahdadou *et al.*, 2025). AI itself cannot be held legally responsible in any governance structure available today, which means the analytical role of the tool and the decision authority of the board must remain separate (Mertens, 2023).

The tool will always end each set of outputs with a statement that *the information delivered here is input for your consideration; the conclusion is up to you.* The tool will never use phrases like we recommend; the board should, or this analysis indicates that this is the right path. The tool is not a recommendation generator.

Strategic framing constraint (DP18)

The outputs from the tool are scoped to board-level strategy concerns such as market positioning, competitive threats, validity of strategic assumptions, environmental change, and the chair determines the boundary of board and management scope when the tool is deployed. As evidenced by Theme 3 and Theme 7, the outwards looking strategic funneling process was uniformly identified by respondents as the responsibility of the board; interviewees made it very clear that the tool should not overstep its bounds into operational issues within the purview of management.

Board strategic influence occurs only in those instances where the board can conduct independent analysis of the strategic environment, not the operational environment (McNulty and Pettigrew, 1999). Consequently, the tool incorporates a scope configuration phase during its implementation where the chair identifies the strategic environments analyzed by the tool and the operational environments excluded from the tool. The tool does not analyze operational data unless requested by the board.

Named internal accountability (DP19)

A named individual is appointed in an information or digital officer capacity as the person who would be responsible for the functioning of the tool, its outputs, and its maintenance. As evident from Theme 8, a number of interviewees stated that a named accountability owner is essential to build trust; their absence was seen to constitute a sufficient condition of non-usage irrespective of the analytical capabilities of the tool. The CEO holds accountability for reviewing and approving tool output before it is delivered to the board.

The lack of accountability as a concrete governance risk arising from the use of AI tools within board-related contexts where there is no specified human accountability owner (Chriki *et al.*, 2025).

Thus, the tool will not be usable if there is no named owner stated in the governance documentation. The name of the owner is specified in the onboarding documents and every output traceable. In other words, the tool cannot be used anonymously.

Board AI literacy as governance infrastructure (DP20)

AI literacy is treated by the organization as one of the key governance conditions required, and this is dealt with through an onboarding process for existing board members and competencies in AI for future recruiting processes in which board members are involved. As evident from Theme 8, several interviewees considered AI literacy as a key precondition for the effective functioning of the AI tool. Lack of AI literacy means that the AI tool's outputs are going to be challenged reflexively, not critically.

Two of the biggest barriers to AI adoption in senior executives were the overconfidence in human decision-making and a sense of control, both of which are mitigated by having knowledge of AI (Marocco, Barbieri and Talamo, 2024). Therefore, this AI governance tool features a two-hour onboarding process for existing board members that allows building basic AI literacy in order to evaluate AI outputs; the onboarding process is also included for each newly recruited board member once deployed.

Governance documentation and audit trail (DP21)

The system will provide a full audit trail of data source access, generated outputs, time of output delivery, completion of CEO curation, and references to outputs made in governance decision records. As mentioned in Theme 7, board members emphasized that responsibility for any decision belongs to them individually; should AI-based analysis help make a governance decision, then the board members need to prove that they have reviewed it objectively. Five legal risks associated with AI usage in the board room are: one being the threat of not being able to apply the Business Judgment Rule because of the lack of documentation of AI impact on a decision. The Business Judgment Rule protects directors against liability as long as their decision is based on proper research (Chriki *et al.*, 2025). The tool will produce a governance decision log for each output cycle listing data sources accessed, confidence scores used, CEO curation where necessary, and date of delivery for each board member.

6 Discussion

This chapter synthesizes the findings of the thesis in relation to its two research questions, locates those findings within the existing body of literature, and reflects their broader significance. Section 6.1 returns to each research question and summarizes the empirical and design answers developed across Chapters 4 and 5. Section 6.2 identifies the theoretical contributions the thesis makes to the intersecting literatures on corporate foresight, board cognition, and AI decision support. Section 6.3 addresses the practical implications of the design framework for boards, tool developers, and governance regulators, including a reflection on the ethical and societal dimensions of deploying AI-based foresight tools in high-stakes governance settings. Section 6.4 sets out the limitations of the study and the conditions under which its findings should be generalized.

6.1 Summary of Key Findings in Relation to Research Questions

The first research question asks: "*What information challenges and foresight gaps do corporate board members experience, and how do these shape the requirements for AI-based decision support?"*

The empirical findings in Chapter 4, supported by the theoretical literature reviewed in Chapter 2, identify four structural information challenges, which the design framework in Chapter 5 then responds to. These challenges recurred consistently across the interviews:: An information architecture problem, in which the filter process of management constrains the epistemic scope of knowledge for the board (Daft and Weick, 1984; Brennan, Kirwan and Redmond, 2016); an episodic governance challenge, whereby the meeting cycle of the board obstructs any form of ongoing monitoring of the environment (Haarhaus and Liening, 2020; Marinković *et al.*, 2022); a deficit within current foresight practice, whereby the board does not perform any structured scanning and scenario building but leaves it up to management to do (Rohrbeck and Gemünden, 2011); and a number of decision support needs, including cross-disciplinary analysis, multilevel communication forms, and pre-board meeting preparation (Simon, 1955; Forbes and Milliken, 1999). All these elements combined point to the gap that exists between the expectations that boards have and what analytical tools they currently have at their disposal.

The second research question asks: "What design and governance principles should govern an AI-based foresight tool for corporate boards to achieve adoption and trust among directors?"

The second research question was concerned with the design and governance principles that should inform the development of an AI foresight tool in order for it to gain adoption and trust by corporate boards. According to the findings discussed in Chapter 4 that lead to design principles in Chapter 5, there are three preconditions that are critical for success: the foresight tool needs to be developed strictly in the sense of augmented intelligence to ensure that decision-making and legal accountability remain within the realm of the board members (Mertens, 2023; Ahdadou, Aajly and Tahrouch, 2024); the adoption of such a tool is based on trust gained through experience of its use, traceable sources and managed change (Marocco, Barbieri and Talamo, 2024; Cooper *et al.*, 2025); and the foresight tool will need to be governed via the principle of accountable governance, auditable governance, and a defined scope (Zeiser, 2024; Chriki *et al.*, 2025).

6.2 Theoretical Contribution

The most important theoretical contribution is the bridge that it builds between two distinct sets of literature that have had little intersection thus far. Foresight studies on corporations have advanced an elaborate framework of the mechanisms by which organizations could systematically enhance their ability to sense and make sense of environmental cues (Daft and Weick, 1984; Choo, 1994; Rohrbeck and Gemünden, 2011). However, this stream of literature has had a corporate practical approach rather than a governance one (Marinković *et al.*, 2022; Engstam *et al.*, 2024). AI-in-Boards literature has focused on the governance challenges presented by AI on the boards of directors, without integrating these issues with foresight research (Mertens, 2023; Pavićević, Haleblian and Keil, 2023; Agnese, Arduino and Di Prisco, 2025; Ahdadou *et al.*, 2025; Ben-Shahar *et al.*, 2025).

The thesis thus acts as a bridge by framing the foresight deficit exhibited by the board as the design problem (through the filters of management, temporal disparity, competitive intelligence and historical scenario optimism), using the foresight literature as a basis for identifying what the AI foresight tool for boards should do. This leads to the design framework where the foresight function feature (Section 5.2.2, Dimension 2) is informed by the foresight literature in a manner which the current AI-in-boards research does not. Design Principle 11, which is about scenario planning and consequence mapping further bridges this gap into the literature on scenario planning, targeting a failure mode of optimism bias and consequence blindness identified in the foresight literature, but not before now articulated in a board-specific context.

The second contribution consists of extending the model of board process, which Forbes and Milliken (1999) created model that has since been applied to other strategic decision contexts, into an AI environment (Pavićević, Haleblian and Keil, 2023). Their three important board process categories effort norms, cognitive conflict, and knowledge utilization are applicable in a pre-AI era. Ben-Shahar et al. (2025) have since extended this model to examine how AI induces epistemic capture, algorithmic groupthink, and effort norm decay. The present thesis extends it in a complementary direction: rather than analyzing how AI threatens these process variables, it derives design requirements for a tool that actively preserves them. Effort norms should not be degraded by an AI-based tool's perceived comprehensiveness (DP14, DP16) (Marocco, Barbieri and Talamo, 2024; Ben-Shahar *et al.*, 2025); cognitive conflict should not be stifled by its consensus-formatting briefings (Forbes and Milliken, 1999; Ben-Shahar *et al.*, 2025); knowledge utilization should not be obstructed by designing it in such a way that it is inaccessible to technologically less competent board members (Barredo Arrieta *et al.*, 2020; Marocco, Barbieri and Talamo, 2024). The design principles for the governance aspect (DP17-DP21) are derived from those potential threats.

Third, the paper expands the automation-augmentation paradox of Raisch and Krakowski (2021) specifically for accountability constraints in governance. Here, it is argued that due to the nature of fiduciary duty and legal requirements of attributability (Mertens, 2023; Zeiser, 2024; Chriki *et al.*, 2025), there is a minimum level of human decision-making authority that cannot be altered according to personal preference (Ahdadou, Aajly and Tahrouch, 2024; Ahdadou *et al.*, 2025). The notion that stage 2 augmentation AI makes decisions that are presented to humans to decide on in a governance context is something not taken into consideration in the original Raisch-Krakowski formulation (Mittelstadt, 2019; Schuett, 2025).

The fourth contribution is the formalization of epistemic independence as a design requirement distinct from explainability. The explainable AI literature establishes that AI outputs should be interpretable to their target audience, supporting verification of individual claims (Barredo Arrieta *et al.*, 2020; Rieger *et al.*, 2025). Epistemic independence operates at a different level: it requires that the tool actively introduces interpretations the board would not otherwise consider, rather than only explaining the interpretation it has produced. Explainability is a property of individual outputs;

epistemic independence is a property of the deliberation environment the tool creates. A fully explainable AI tool can still produce epistemic capture if every output it explains points in the same direction (Ben-Shahar *et al.*, 2025); epistemic independence is the design response to that risk, embedded in DP9 (challenging perspectives), DP11 (multiple scenarios including pessimistic), and the deployment principles that preserve the conditions for divergent reading (DP14, DP17).

6.3 Practical Implications

In relation to *corporate boards*, the twenty-one principles become a set of systematic criteria for the evaluation and procurement of AI foresight technology. The most valuable first application is for corporate boards to conduct an initial systematic analysis of their information environment, asking questions such as what information signals their management filters out, how their rhythm of decision-making stacks up against the rate of disruption in their market sector, and whether any competitive intelligence architecture already exists (Daft and Weick, 1984). While individual psychological factors such as overconfidence and desire for control are recognized barriers (Marocco, Barbieri and Talamo, 2024) adoption challenges at board level are fundamentally organizational as they require structural change management, AI literacy programs, and embedded governance protocols, not just individual willingness.

For the *developer* segment, the framework provides a product specification based on those dimensions which are most important to the user. The competitive intelligence feature is the clearest indication of product-market fit, being the key application identified among participants as most valuable. The adjacent industry scanning capability is an important differentiating factor that existing competitive intelligence solutions ignore; the very nature of such a solution based on its own industry classification reinforces the conditioned viewing problem it was meant to address (Daft and Weick, 1984; Robinson and Simmons, 2018). Design Principle 11, which is the ability to build scenarios and map consequences as mentioned before is another capability offering a unique commercial value proposition addressing a failure mode consistently identified in the foresight literature: scenario planning that produces probability-weighted outcomes without mapping the consequence cascades that give those scenarios strategic meaning for boards (Marinković *et al.*, 2022; Grove, Clouse and Xu, 2023). The change management protocol is an essential part of any product specification; highly effective tools lacking a clear implementation strategy will face the same adoption cliff mentioned by one innovation in the empirical study (Davis, 1989; Venkatesh *et al.*, 2003; Marocco, Barbieri and Talamo, 2024).

The *governance standards* of organizations and regulators, the filter effect discovered by the management team, indicates that the prevailing governance models tend to overlook the information environment aspect of effective board functions (Daft and Weick, 1984; Brennan, Kirwan and Redmond, 2016). The EU AI Act introduces a new board-level accountability responsibility, namely the need for directors to determine whether their organizations have correctly classified, managed, and documented their AI systems (Zhao, 2024; *Regulation (EU) 2024/1689 (Artificial Intelligence Act)*, 2024). Such a need is yet to be considered sufficiently by most boardrooms (Evgeniou *et al.*, 2025). AI literacy skills, together with knowledge of the EU AI Act, should be introduced as threshold competencies at the board level just like financial literacy skills (Marocco, Barbieri and Talamo, 2024; Evgeniou *et al.*, 2025).

6.3.1 Ethical and Societal Reflections

The design of AI-based foresight tools for corporate boards raises ethical and societal questions that extend beyond the governance of individual organizations. This section reflects on five dimensions: who benefits and who is at risk from board-level AI, accountability when AI influences governance decisions, the risk of deepening rather than reducing information asymmetries, the potential for AI

to improve attention to sustainability signals, and the ethics of designing powerful analytical tools for already-powerful actors.

Foresight technologies in AI created according to the framework of this thesis are far from being neutral. They are meant for use in corporate boards, which are powerful economic entities influencing their employees, suppliers, and communities and other stakeholders involved in different ways. The ones who *benefit* directly from improved foresight skills in boards are mainly the corporations themselves and their shareholders. However, the potential *negative consequences* could have wider effects and influence many people. Directors who rely on foresight created by an AI tool but do not verify it properly may make certain decisions affecting large numbers of stakeholders while these stakeholders cannot see what is going on in the background due to the lack of transparency. The concentration of advanced AI technologies in powerful entities might lead to greater inequality rather than solving social problems when no public control or transparency is available (Mittelstadt, 2019). The design principles proposed in this thesis mitigate this problem by including source transparency (DP12), audit trails (DP21), and augmented intelligence constraint (DP17). Still, this does not answer the question of whether there is some need for external disclosure of the results generated in board AI (Papagiannidis, Mikalef and Conboy, 2025).

Accountability emerges as the most pressing ethical dimension from a purely legal perspective within the context of this thesis. When the board takes a decision based on AI analysis, the issue of whose responsibility that decision rests and under what proof becomes a governance as well as a legal problem. The Business Judgment Rule, which ensures the legal immunity of directors in case decisions are made based on reasonable information and in good faith, was formulated in a time where the inputs for such decisions were curated by people and verifiable (Chriki et al., 2025). With the use of AI, however, there is a gap in attribution because it becomes difficult to prove that the directors had considered, debated, and independently judged the analysis presented to them, rather than accepted it unquestioningly (Zeiser, 2024). The requirement of an audit trail, named ownership, and augmented-intelligence approach in the governance and accountability aspect of this framework (DP17-DP21) tries to deal with this concern. From an ethical perspective, however, it goes far beyond a mere legal requirement; accountability to other stakeholders mandates that there should be an audit trail for all AI-driven decisions that have a consequence (Mertens, 2023).

Management Filtering is one of the four major structural challenges to be considered as identified in the thesis. Using foresight technology through AI can be seen as the solution to this problem since it will provide independent information source for the boards. However, there exists another risk of generating *misalignment* as the tool can create a different perspective for the boards compared with the managers who are supposed to use the tool (Brennan, Kirwan and Redmond, 2016). In effect, while the use of the foresight tools was meant to correct some of the structural problems of the boards, it may end up creating new unintended effects and consequences that are not yet empirically explored in the literature. The current imbalance of information that generates dysfunctional accountability relationships is expected to become worse when an AI-based tool becomes an exclusive property of the board of directors (Brennan, Kirwan and Redmond, 2016). Ethical concern regarding the use of these kinds of technologies should include the formulation of governance policy on how the AI-driven board analysis should relate to management reports.

As per the design of the tool, emphasis is placed on the scanning of competitive, regulatory, and strategic environmental signals. The proposed framework does not make scanning for sustainability or ESG foresight a mandatory requirement; however, in principle, DP8 (weak signal detection and adjacent industry scanning) could potentially be modified to incorporate climate risk signals, sustainability trends in regulatory affairs, and social transition signals. This is especially important since sustainability signals, which include such factors as climate risks, ethical supply chain issues, and social transitions, form some of the most disruptive and fast-moving signals in the present-day strategic environment (The World Economic Forum and OECD, 2025). In this context, future

iterations of the proposed framework must examine whether the sustainability signals require mandatory scanning, along with competitive and regulatory signals, taking into consideration the increasing need for board level sustainability disclosures due to sustainability reporting obligations at the EU level (European Parliament and Council, 2022)

A fifth dimension is harder to reduce to a single concrete risk: the underlying ethics of designing analytical tools that strengthen the capacity of already-powerful actors. Design Science Research, the methodology used here, proceeds from the assumption that the artefact produced will yield social benefit (Hevner *et al.*, 2004; Gregor and Hevner, 2013). That assumption is worth interrogating when the intended users are corporate boards. The twenty-one principles in this thesis are oriented toward making boards better informed, less captured by management framing, and more resilient against AI-induced epistemic narrowing. These are improvements in governance quality, but governance quality is not value-neutral when seen from outside the boardroom. If tools that are going to enhance the analytical capacity of corporate boards fail to concurrently enhance transparency of board decision-making process to outside parties, then we would be facing the problem of 'ethical gap' (Mittelstadt, 2019). This framework helps to bridge the gap to such an extent as it maintains the conditions necessary for human accountability and traceability through its governance-related design principles DP17-DP21. However, the mechanisms of internal accountability are enough in face of social impacts of board decision-making process remains open to discussion (Papagiannidis, Mikalef and Conboy, 2025; Schuett, 2025).

6.4 Limitations

Size of Sample and Diversity. Insights gained from the study are based on semi-structured interviews conducted with ten individuals. Despite having theoretical content to back up specific constructs, and despite the increased sample size leading to a significant convergence of themes, the sample size can by no means be taken as statistically representative of the population under investigation (the population being corporate boards in Sweden). Participants involved in the research are mostly senior individuals, mainly recruited using Nordic network connections, which have an innovative tendency. There is a lot of diversity in terms of sectors investigated, namely fintech, energy, technology, pharma/biotech, medtech, and digital health; however; however, there is also a significant concentration of people occupying chair positions in the corporate sector with considerable board-related experience.

Single governance setting. The research took place exclusively in the corporate governance setting of Nordics, more specifically Swedish, that is marked by vigorous boards, co-determination practices, and fairly high levels of institutional trust. These factors influence the character of relations between the board and management, the voluntary adoption of AI, and the legal environment of fiduciary responsibility. The principles of the designed framework can be applied in other governance settings similar to the one in question; whether or not it can be done so in relation to the US single tier board structure, the Japanese keiretsu governance, or emerging markets setting needs further empirical examination.

Little practitioner recognition of fiduciary and governance risks. The empirical data is silent on certain governance risks that the literature treats as central. Few interviewees raised fiduciary or attributability concerns spontaneously, with the EU AI Act mentioned by only one participant. Mertens's (2023) discovery risk, Chriki et al.'s (2025) fiduciary risk, and Zeiser's (2024) attributability risk did not surface as practitioner-articulated concerns during the interviews. This silence does not mean that the risks do not exist in principle. Consequentially, the governance and accountability principles in Dimension 4 are primarily grounded in the literature rather than in interview data and remain theoretically derived rather than practitioner-validated. As directors gain practical exposure to AI tools and as the general understanding of AI governance risks evolves,

future empirical work may produce different results. Research conducted in such a mature landscape will be necessary to see if practitioner awareness aligns with the literature once these risks transition from hypothetical to practical.

Absence of prototype validation. Design principles cannot be verified through tool usage in this thesis, which is a necessity according to design research (Hevner et al., 2004). Instead, the principles are developed on the basis of both theoretical deductions and practitioner experience, which constitutes a sound source for design knowledge; yet this approach does not amount to empirical verification of tool effectiveness. These design principles can be falsified through empirical testing of a tool designed on the basis of these principles: the effectiveness of such a tool could be assessed through adoption rates, trust calibration paths, and deliberation quality.

7 Conclusions

The purpose of this thesis was to develop a design framework for an AI-based decision support system that enhances board-level foresight and strategic environmental scanning. The literature establishes that no such framework currently exists despite a growing case for board-level AI. Four structural information challenges, identified consistently across the empirical data and supported by the theoretical literature, motivate the framework: a management information filter that constrains the board's epistemic horizon by controlling what information reaches directors and how it is framed (Daft and Weick, 1984; Brennan, Kirwan and Redmond, 2016); a temporal mismatch between the episodic cadence of board governance and the pace of strategic disruption in the external environment (Haarhaus and Liening, 2020; Marinković *et al.*, 2022); a deficit in structured foresight practice at board level, whereby scenario planning is left to management, tends toward optimism bias, and fails to map consequence cascades that would give boards an independent view of alternative futures (Rohrbeck and Gemünden, 2011; Grove, Clouse and Xu, 2023); and a competitive intelligence gap, whereby boards lack access to external scanning that is independent of the management channel and extends beyond the firm's immediate competitive set to adjacent industries and macro-environmental signals (Daft and Weick, 1984; Robinson and Simmons, 2018).

The thesis addressed this gap through a Design Science Research approach (Hevner *et al.*, 2004), integrating a seven-stream literature review with qualitative empirical evidence from ten semi-structured interviews with senior board practitioners and one expert advisory session. The thematic analysis produced eight themes organized across three layers (context, capability, adoption). The synthesis of theoretical and empirical material resulted in a design framework including twenty-one principles categorized into four dimensions: cognitive augmentation requirements (DP1-DP6), foresight function requirements (DP7-DP11), trust and adoption requirements (DP12-DP16), and governance and accountability requirements (DP17-DP21).

In answer to RQ1, four structural information challenges were identified that consistently shape what boards require from AI-based decision support. For RQ2, twenty-one design and governance principles across four dimensions specify the conditions under which such a tool can be adopted and trusted without undermining directors' fiduciary accountability. The argument underlying this thesis is that AI has the potential to overcome all four structural problems related to information, provided it is developed while considering three constraints: the tool must preserve the deliberative conditions (cognitive conflict, knowledge utilization, effort norms) that make boards effective decision-making groups (Forbes and Milliken, 1999); it must be deployed through a structured change management protocol (Theme 7) that builds AI literacy, integrates the tool into the governance agenda, and develops trust through repeated use (Ben-Shahar *et al.*, 2025), as deployment of an advanced AI tool without proper change management is likely to fail (Marocco, Barbieri and Talamo, 2024); and it must be governed in a way that preserves directors' fiduciary accountability (Mertens, 2023; Ahdadou, Aajly and Tahrouch, 2025; Chriki *et al.*, 2025).

The framework is presented as a living artefact rather than a static specification. The legal, regulatory, and technological context in which board-level AI operates is changing rapidly: the EU AI Act's enforcement obligations are still being phased in, AI capabilities continue to evolve, and Swedish corporate governance practice has yet to consolidate around the question of AI use at board level. Specific design principles may require revision as these conditions develop; in particular, the augmented-intelligence constraint (DP17), the public-data default (DP15), and the documentation requirements (DP21) are likely to be the principles most sensitive to regulatory and technological change. The framework's four dimensions, however, represent more enduring categories of design concern that should remain analytically useful even where individual principles are revised.

7.1 Future Work

Prototype development and evaluation. The most direct extension of this thesis is the construction and empirical evaluation of a tool built according to the twenty-one design principles. This represents the natural continuation of the design science cycle, in which the design knowledge developed here is operationalized, observed in use, and iteratively refined (Hevner *et al.*, 2004). Evaluation criteria derived from the framework include adoption rate, trust calibration trajectory, deliberation quality before and after tool introduction, and compliance with the governance documentation requirements of DP21. Such evaluation would also enable empirical testing of the framework's central claims that cannot be validated on the basis of design knowledge alone.

Epistemic capture in deployed board AI systems. When company boards rely heavily on AI to help them make decisions, they risk losing their ability to think independently. The theoretical model of epistemic capture by Ben-Shahar et al. (2025), which includes algorithmic groupthink, pseudo-consensus, and effort norm decay, makes certain predictions about failures that are not yet empirically verifiable since board-level AI foresight tool usage is at an early stage. A longitudinal study on the influence of AI tools on board deliberation quality, cognitive diversity, and decision independence could provide the empirical validation missing from epistemic independence. Attention should focus particularly on ecosystem-wide convergence, where boards working with similar AI tools have overlapping cognitive biases that only the ecosystem as a whole can recognize.

Replication across other governance contexts. This approach is developed for and has the closest applicability in the Nordic / European corporate governance context, marked by active boards, co-determination principles, and higher institutional trust levels. Its replicability within the US one-tier board context, Japanese keiretsu-style corporate governance, and in emerging economies' corporate governance frameworks would determine its broad applicability or point towards the adaptations needed for each respective context. Differences between jurisdictions in the structure of fiduciary responsibility systems, principles guiding the manager-board relationship, and regulatory regimes of AI technology could impact the principles to pay attention to and implement.

AI Literacy Among Directors as a Quantifiable Measure of Governance. Design Principle 20 highlights AI literacy among directors as governance infrastructure akin to financial literacy; however, no information is provided regarding what constitutes board AI literacy at various levels of sophistication in the deployment of AI, how to assess it, and how to foster it. Research on board AI literacy as a quantifiable governance criterion, akin to the frameworks of financial literacy created for audit committee governance, would transform a design principle into a tangible governance framework, especially considering that the oversight measures required by the EU AI Act would presuppose some measure of AI literacy among board members.

Towards Greater Sample Diversity in Board-Level AI Research. The current study used ten board members, a sample which was enough to achieve the core themes, although it represents only a small fraction of the actual corporate governance settings. If there had been an opportunity to conduct interviews with a much larger sample, involving directors from various national corporate governance backgrounds, public versus privately held companies, large versus small organizations, and different levels of board digitalization maturity, the results could have looked differently in many aspects. The use of AI technologies by directors in their daily activities may lead them to identify the requirements for design that can be more technically detailed, whereas other requirements are based on assumptions of trust in some situations. Moreover, interviewing a wider range of board members may allow discovering some foresight strategies or factors impeding the implementation of foresight strategies which have not been revealed in the current sample due to the sectoral or cultural specificity of the board members' experiences.

Another potential future direction involves the *homogeneity of the sample* across two factors which the present study did not take into account. Namely, there is a great variation in the sample's AI literacy, ranging from directors with extensive experience in using AI solutions in their organizations to members of boards who had come across AI systems only in passing. Although such diversity provided good insights into varying attitudes towards adoption of the technology, it also suggests that the design requirements outlined may represent a certain middle ground between two very different sets of preferences rather than true requirements of either group. Further research can investigate this question by sampling a narrower segment of respondents: those who already have a certain level of AI literacy and use AI solutions regularly in their professional environment. Such a sample would allow the study to explore whether higher literacy leads to significantly different requirements, higher trust levels, or greater willingness to use the 'opponent' and 'contradiction' features, which were introduced with some reservations in the present research.

Second, the current study had nine industries within its scope, providing scope for broad applicability across multiple contexts while making it challenging to narrow down sector-specific design considerations to an appropriate degree of detail. A study conducted within the *context of one industry/market sector*, such as the financial services industry, which is known to have high pressure with regard to AI governance from regulators and whose boards have above-average levels of data literacy, could provide a more detailed understanding of foresight tool requirements in practice. Moreover, this type of research could confirm whether the management filter, temporal mismatch, competitive intelligence deficit, and structural optimism identified in this study as the four key issues apply equally well in an industry that receives more structured external intelligence at the board level than the average Nordic setting.

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Appendix A: Search Strings

String 1 - KTH Library AI-generated string (refined):

((AI support) AND (foresight for board members)) OR (artificial intelligence impact on board of directors' strategic foresight) OR (board of directors AI foresight) OR ((AI) OR (machine learning AND board governance foresight)) OR ((AI-driven foresight) AND (corporate board decision-making)) OR (("AI support" AND (foresight) OR (future planning) AND (board members) OR (board of directors))) OR (artificial intelligence impact foresight corporate governance) OR (((AI) OR (artificial intelligence) AND (foresight) OR (strategic foresight) AND (board) OR (directors))) OR (("board of directors" AND "strategic foresight" AND (AI) OR ("artificial intelligence"))) OR (corporate foresight AI leadership decision-making) OR (AI support and impact on foresight for board members/board of directors)

String 2 - AI adoption and governance:

> (AI OR Artificial intelligence) AND (adoption barrier*) AND (executive OR governance OR board*) NOT (supply chain OR clinical OR healthcare)
>
> Filter: 2020–2026, English, peer-reviewed, articles
>
> Results: 256

String 3 - AI, boards, and decision-making:

> (Artificial intelligence OR AI) AND (Board*) AND (decision) AND ((Scenario OR vision OR future) OR (social) OR (legal OR regulat*) OR (ethic*)) NOT (healthcare OR clinical)
>
> Filter: 2020–2026, English, articles - artificial intelligence, board of directors, decision making, ethics, corporate governance, leadership
>
> Results: 412

Appendix B: Inclusion and Exclusion Criteria

Papers were evaluated against the following criteria before being admitted to the final selection:

Criterion	Inclusion	Exclusion
Language	English	Non-English
Publication type	Peer-reviewed articles; grey literature where directly relevant to AI foresight	Conference abstracts without full text
Relevance	Directly addresses one of the seven theoretical streams	Tangentially related only
Recency (AI literature)	2015-2026 prioritized	Pre-2015 AI literature unless foundational
Recency (governance/foresight literature)	No hard cutoff; seminal works included regardless of date	-x-
Scope	Corporate boards, strategic foresight, bounded rationality, AI decision support, technology adoption, XAI, board cognition	Clinical, healthcare, supply chain, or other non-governance AI applications

Appendix C: Interview Guide

Interview Guide 1st Iteration

This appendix presents the interview questions and scenario used across all interview sessions. Interviewer instructions, probes, follow-up prompts, and reflection templates have been omitted. The questions are presented in the order they appeared in the final version of the guide.

Section A: Warm-Up

Q1. Could you briefly describe your current board role(s) and how long you have been involved in board work?

Q2. How would you describe your current relationship with data and technology in the boardroom?

Section B: Information and Decision-Making at Board Level

Q3. When your board faces a major strategic decision, how does the relevant information typically reach you?

Q4. When you prepare for a board meeting, do you ever feel that the information you receive gives you a complete enough picture to make the decision with confidence?

Q5. Beyond what management provides, do you actively seek additional information before board meetings? If so, where?

Q6. In your experience, are there types of strategic issues that boards tend to overlook or not give enough attention to? If so, why do you think that happens?

Section C: Strategic Foresight and Environmental Scanning

Q7. How does your board currently work with long-term strategic issues for example, emerging market shifts, technological disruption, or regulatory changes that may not be urgent today but could be in the future?

Q8. When someone raises a strategic concern that challenges the direction management has proposed, how does the board typically respond?

Q9. If you think about the biggest strategic surprises your board has faced in recent years something you did not see coming, what happened, and why do you think it was missed?

Q10. If you could change one thing about how your board handles strategic foresight and forward-looking analysis, what would it be?

Section D: Attitudes Toward AI-Based Decision Support

Q11. What is your general view on AI being used to support board-level decision-making for example by summarizing information, surfacing external signals, or highlighting alternative perspectives?

Q12. If an AI-based tool could help your board with strategic foresight, what would be the most valuable thing it could do?

Q13. What would definitely NOT work? What should an AI tool for boards absolutely avoid doing?

Section E: Trust, Transparency, and Adoption Conditions

Q14. When you receive a piece of analysis or an insight whether from a consultant, an internal report, or a tool, what makes you trust it?

Q15. If an AI tool generated an insight for your board for example, flagging that a competitor is shifting strategy based on external data analysis, what would you need to see to take it seriously?

Q16. How should AI-generated information be presented to a board in order to be useful in practice?

Q17. What format would work for directors who have limited time and high standards for quality?

Q18. Who do you think such a tool should primarily serve in practice: individual board members, the chair, a committee, a board secretary, or the board collectively?

Q19. What would make you personally willing or unwilling to use an AI-based tool as part of your board preparation?

Section F: Governance and Accountability

Q20. As a board member, you carry legal and fiduciary responsibilities. How does that affect your willingness to rely on tools or analyses that you did not produce yourself?

Q21. If a board used AI-generated insights as part of its decision-making and the decision later turned out badly, who should be accountable?

Q22. Are there types of board decisions where AI support would be clearly appropriate, and types where it would be clearly inappropriate? Where would you draw the line?

Section G: Design Probe - Scenario Reaction

Participants were presented with the following scenario before being asked the questions below.

Scenario

Imagine your board is preparing for a quarterly strategy meeting. Two weeks before the meeting, an AI-based tool scans publicly available data (industry reports, regulatory filings, patent databases, news sources, and macroeconomic indicators) and produces a short briefing document for each board member.

The briefing highlights three things: (1) two emerging market signals that are not covered in management's board materials, (2) a potential risk scenario where a competitor's recent moves could affect your company's market position within 12–18 months, and (3) a data point that contradicts an assumption in management's strategic plan.

Each item includes a plain-language summary, a confidence rating (high / medium / low), and clickable links to the original sources so you can verify the claims yourself. The tool does not make recommendations. It presents information and flags what it thinks the board should be aware of.

Q23. What is your first reaction to this scenario? Would something like this be useful to you?

Q24. What would you want to change about how this tool works, what is missing, what is unnecessary, or what would make it more useful?

Q25. What concerns would you have about receiving this kind of briefing?

Q26. If you imagine yourself receiving this before a board meeting, would it change how you prepare or how you participate in the discussion?

Q27. If a board were to start using a tool like this seriously, what would need to change in the board's way of working for it to be used well?

Section H: Closing

Q28. Is there anything we have not covered today that you think is important for us to understand about how boards work with information and how technology might fit into that?

Interview Guide Final Iteration

This appendix presents the interview questions and scenario as used in the *final version of the guide*. Interviewer instructions, probes, follow-up prompts, timing notes, and administrative materials have been omitted.

A. Introduction & Background Questions

Q1. Could you briefly describe your background and current board roles?

How many boards are you on, and do they span different sectors?

Have you been engaged in other boards previously? How many in total?

Q2. What kind of strategic decisions are your boards most actively engaged in right now?

Do the strategic challenges tend to differ across boards, or are there any common themes?

Q3. How does the relevant information typically reach you?

B. Information Challenges

Q4. Ask whether boards ever look into adjacent industries, not just direct competitors.

Q5. Some issues become important over time instead of right away. How does your board consider long-term perspectives today, if at all?

Q6. In your experience, have there been situations where issues like these only received real attention once they had become more visible or more urgent?

Q7. Imagine your board receives signals that an environmental development, a regulatory shift, a technology disruption, or a move by a competitor, may be challenging your current strategy. If you wanted to understand what that actually means for your strategic position: what questions would you ask yourself first? Where would you start looking?

C. Gap & Transition before Scenario

Q8. If you think about your board's work today, is there any kind of information, perspective, or early signal that you wish reached the board earlier or more clearly than it does now?

Q9. If you think about how strategic information reaches the board and is discussed during meetings, is there any part of that process you think could work better or be improved?

Q10. Is the challenge about the information that reaches the board, or about how it is discussed and used once it is there?

Q11. Do you believe strategy work will change in any way the upcoming 2 years? In what way?

How can AI support that change?

D. Scenario

D1. Partial Scenario

SCENARIO

Imagine your board is preparing for a strategy meeting. About XX [adjust time to how they receive it now] before the meeting, an AI-based tool produces a short briefing document for each board member based on publicly available information such as industry reports, regulatory updates, news sources, market developments, and competitor activity.

D1-Q1. Based on what you have just described about how your board works, what kind of information or signals would you have wanted highlighted in something like that?

D1-Q2. If you received something like this before a meeting, what would make you actually engage with it rather than set it aside?

D1-Q3. What would make you trust the information in it or be skeptical?

D2. Full Scenario

SCENARIO (FULL Scenario)

The briefing highlights a small number of issues that the tool considers strategically relevant for the company:

- *emerging changes in the external environment*
- *possible risks (e.g. a potential risk scenario where a competitor's recent moves could affect your company's market position within 12 to 18 months)*
- *information that may challenge current assumptions in the strategic plan*

Each item includes a short summary, a confidence rating (high / medium / low), and a link to the original source so you can review it yourself. The tool does not make decisions or recommendations. Its role is to synthesize information and flag issues that the board may need to be aware of before the meeting.

D2-Q1. What is your first reaction to a tool like this? Can you see something like this being useful in your board context?

D2-Q2. What would make a briefing like this practically useful in your board work?

D2-Q3. If all board members were to receive this briefing, how do you think it should enter the board's discussion?

D2-Q4. Imagine that the tool repeatedly highlighted one area in an inaccurate way, and nobody noticed it at first. What kind of oversight or review process would be needed for something like that to be caught in time?

D2-Q5. If your board were to adopt a tool like this, who inside the organization should have responsibility for it overall? E.g. the chair, the company secretary, a committee, or someone in management?

E. Design Principles

Q12. Thinking about everything we have discussed so far, what do you think is most important for a tool like this to get right in order to actually be valuable in a board context? E.g. information it brings forward, how the information is presented, how it fits into board work, or how trustworthy it feels?

Q13. And on the other hand, what would be most important to avoid, so that a tool like this does not create new problems in board work?

E.g. over-reliance, too much information, poor fit with the board's process?

F. Trust & Adoption

Q14. Thinking about everything we have discussed, what do you think matters most for a board to feel comfortable trusting a tool like this in practice?

Q15. On a personal level, what would make you hesitate most in using a tool like this yourself?

Q16. Can you think of any situation where you believe that the board should not rely on AI-supported information, even if it seems accurate?

G. Closing

Q17. If you had to choose one thing overall, what do you think would matter most for a tool like this to work well in a board context?

Extra Questions (if time remained)

E1. Have you or any of your boards tried AI or data-analytics tools for strategic analysis or horizon scanning? If yes, what was your initial reaction?

E2. Are there particular types of strategic decisions, e.g. market entry where AI-assisted foresight would add the most value? And anywhere it clearly would not help?

E3. What do you think would need to change for tools like this to become a more normal part of board work, rather than something only a few boards would use?

Do you think the biggest barrier is cultural, generational, liability, or something else?

E4. Is there anything we have not covered that you think is important, about AI in boardrooms, or about strategic foresight more broadly

Appendix D: Thematic Analysis Coding

Reference for Interviewee IDs:

Informant	Role / Background	Sector	Board Experience	AI Familiarity
I1	Fintech Chairman	Fintech	10+ years	High (active user)
I2	Board Member	Energy	7 years	Moderate
I3	Former Group CEO	B2B Tech/Digital	20+ years	Moderate
I4	Former Head of R&D, International board member	Pharma / Biotech	30+ years	Low
I5	Former CFO	Tech	9 years	High (active user)
I6	Former CEO	MedTech	20+ years	Moderate
I7	Former CEO / Chairman	Pharma	20+ years	Low
I8	Board Member	Fintech	2 years	High (active user)
I9	Head of Innovation	Digital Health	10 years	High (specialist)
I10	Professional board member	Tech	20+ years	High (active user)
I11	Expert Advisory Session	Foresight Methodology	N/A (expert advisory)	High (specialist)

Thematic Analysis Table:

Thesis theme	Code label	Code Description	Interviewees IDs
Theme 1: Information Architecture Problem	Information Quality Deficiency	Board material are described as unclear, not strategically framed, or insufficient in analytical depth	I3, I4, I10
	Company Size as Quality Driver	Larger and more mature companies produce better board material; SMEs and smaller organizations struggle	I3, I4, I6, I8
	Fragmented and Delayed Delivery	Material arrives piecemeal and sometimes late, hampering individual preparation before meetings	I1, I2, I3, I4, I6
	Material Volume and Overload	Board material is often excessive in volume (long decks, in some cases several hundred slides or pages), creating information	I1, I2, I3, I4, I5, I6, I10

		overload that hampers focus and efficient preparation	
	CEO as Information Gatekeeper	Management produces and the CEO confirms the information reaching the board, controlling not only what the board sees but how it is framed. This provides a single point of coordination and editorial accountability (avoiding executives being blindsided), but also creates a structural dependency and information-asymmetry risk.	I1, I2, I3, I4, I5, I6, I7, I10
	Uneven Individual Preparedness	The degree of independent supplementation, and therefore how well-prepared each director is, varies with individual time, networks, and prior experience, producing unevenness in board preparedness.	I1, I3, I4, I5, I7, I10
	AI and External Supplementation	Board members already use ChatGPT, Copilot, and Perplexity individually to fill information gaps	I1, I2, I3, I5, I6, I8, I9, I10
Theme 2: Episodic Governance Trap	Frequency and Timing of Formal Meetings	Most boards hold strategic sessions once or twice per year, tied to the annual general meeting and annual planning cycle	I1, I2, I3, I4, I5, I6, I7, I8, I9, I10
	Strategy Stability versus Adaptability	Tension between keeping strategy stable over time and adapting to rapid disruptive environmental changes	I1, I2, 14, I5, I6, I7, I8, I9, I10
	Reactive Updates Between Formal Cycles	Urgent external events trigger ad hoc strategic discussions	I2, I3, I5, I7, I8, I10
	Strategic Gaps Between Cycles	The episodic meeting rhythm leaves the board exposed to environmental shifts between sessions, creating gaps in strategic awareness and decision-making between annual/biannual meetings.	I1, I2, I7, I9
Theme 3: Strategic Days as Integration Point	Structure and Format of Strategic Days	Strategic days follow recognizable but highly variable formats across organizations and sectors	I1, I2, I3, I5, I6, I8, I10
	Management-Driven, Board-Endorsed	Strategy substance is developed by management and brought to the board for challenge and endorsement	I1, I2, I5, I6, I7
	Environmental Scanning as Input to Strategic Days	External analysis of macro trends, competitor moves, and technology developments feeds the strategic day funnel	I1, I2, I3, I4, I5, I6, I8, I9, I10
	Strategy to Financial Planning Cascade	Strategic decisions from strategic days cascade into financial planning and target-setting processes	I1, I2, I4, I5
	Quality of	Value of strategic days lies in the quality of	I3, I10

	Collective Conversation as Enabler	collective thinking and deliberation, not only in the outputs produced	
	Company Size and Planning Depth	Larger and more mature companies have more formal, long-horizon strategic processes than smaller organizations	I4, I9, I10
	External Perspectives to Counter Narrow Focus	Boards deliberately bring in outside voices (e.g., external experts, students, customers) to disrupt an internally narrow point of view and broaden strategic discussion	I1, I4, I11
Theme 4: Current Foresight Practices and Gaps	Growing Complexity as Foresight Driver	AI, sustainability, geopolitical instability, and tariff shifts create unprecedented strategic uncertainty for boards	I1, I2, I5, I7, I10
	Foresight as Individual, not Collective	Where foresight happens it is informal and individual; environmental awareness is not synthesized across the board, so members hold different readings rather than a shared forward-looking picture.	I5, I8
	Scenario Building as Core Foresight Practice	Scenario building is the primary foresight method identified by interviewees, exploring multiple possible futures	I4, I5, I6, I7, I8, I10
	Limitations of Historical Scenario Planning	Traditional methods were systematically optimistic, infrequent, and miss risk consequences	I2, I7, I9
	AI as Scenario Enhancement Tool	AI potential to improve scenario depth and speed	I1, I6, I7, I8
	Foresight as Information Exploration	Foresight is exploratory and distinct from decision-making, which positions it as lower-oversight and lower-risk	I3, I5, I10
	Early Signal Detection and Missed Opportunities	Boards miss early signals and respond reactively to strategic developments	I3, I6, I7, I9, I10
	Foresight as Organizational Mindset	Effective foresight requires a company-wide forward-looking mindset, not only a board-level analytical tool	I1, I5
	Opposed views of Foresight Responsibility	Disagreement on where foresight responsibility should sit: (a) within management, with the board reviewing outputs; (b) the board catalyzing foresight within management without doing it itself; or (c) independent board-level foresight as a precondition for challenging management.	I1, I3, I5, I8, I10
	Continuous versus Periodic Foresight	Tension between the annual foresight exercise model and the need for continuous	I1, I2, I3, I4, I6, I7, I9

		real-time environmental monitoring	
Theme 5: Decision Support Requirements	AI as Cross-Domain Strategic Intelligence Aggregator	AI potential to synthesizes signals across multiple external domains into contextualized strategic insights for the board	I1, I2, I5, I7
	AI-Enhanced Scenario Building	AI have potential to improve speed, and quality relative to manual approaches	I2, I5, I7
	Interactive Dialogue as Key AI Value	Conversational and iterative AI interaction adds value	I1, I2, I4, I5, I9
	Layered Information Access	AI should enable multiple levels of depth, from executive summary to detailed drill-down on demand	I3, I4, I6,
	Tool Use-Case: Preparation vs In-Meeting	AI is most valuable in the preparation phase before meetings rather than as a real-time tool during meetings	I1, I2, I5, I3
	Format and Presentation Quality	Clean, pedagogically clear, and simple presentation formats preferred over dense data outputs	I1, I2, I3, I5, I7
	Output/Scenario Overload Risk	Too many tool-generated scenarios or constant what-if analysis risks scattering and overwhelming board discussion, so output must be curated and prioritized	I2, I3, I5, I8, I10
	Customization for Industry and Company Context	AI tools need tailoring to specific industries, company sizes, and strategic contexts to be useful	I1, I3, I5, I7, I8
	AI as Independent Strategic Challenger	AI can actively surface evidence that challenges and stress-tests current board strategic assumptions	I1, I2
Theme 6: Trust Architecture for Board-Level Tools	AI Accuracy Threshold	AI output does not need to be perfectly accurate to be useful	I1, I2, I7, I8, I9
	Hallucination and Accuracy as Trust Barrier	AI inaccuracy and hallucination is the most consistently cited reason for board member hesitation	I4, I5, I6, I7, I9
	Trust Builds Gradually Through Experience	Trust in AI tools develops incrementally through demonstrated accuracy over repeated use cycles	I1, I4, I5,
	Information Security as Adoption Barrier	Data leakage and concerns about cloud-based tools processing confidential strategic data are significant barriers	I1, I5, I8, I10
	Cross-Checking as Standard Practice	Board members consistently cross-check outputs across multiple sources as standard preparation behavior	I1, I2, I4, I7, I9

	Confidence Rating and Source Transparency	Confidence ratings and direct source linking are valued trust-building features of tool output	I1, I2, I4, I5, I7, I9, I10
Theme 7: Human Judgment as Non-Negotiable	Human Intuition and Context Beyond the Tool	AI is a supporting tool that cannot replace human board judgment or accountability. Complex strategic choices draw on accumulated experience and intuition ('gut feeling'), and on informal/undocumented signals the tool cannot access; AI provides only 'a piece of the puzzle	I2, I9, I10
	Critical Thinking as Non-Negotiable Safeguard	Board members must maintain their own judgment and critical analysis regardless of AI output quality. Accountability for AI tools are seen as either needed to maintain trust and enable proper oversight, or not needed as board members already holds accountability	I6, I8, I9
	Legal Accountability as Human Differentiator	AI cannot hold legal or fiduciary responsibility. Clear boundary what AI can and cannot do.	I1, I2, I3, I5, I6, I7, I9
Theme 8: Conditions for Adoption	Named Internal Ownership	A named individual must be accountable for tool operation, outputs, and oversight	I1, I5, I6, I8, I9, I10.
	CEO/Named Review Before Board Delivery	CEO or a named reviewer should approve AI-generated output before it reaches the board as official material	I2, I3, I5, I6, I7, I8
	Structured Deployment with Change Management	Tool introduction requires structured onboarding, calendar integration, and a defined pilot phase	I1, I4, I5, I9
	AI as Efficiency and Time-Saving Tool	AI is valued primarily for time savings and the accessibility of synthesized information	I2, I3, I4, I5, I7, I9, I10
	AI Literacy as a Baseline Adoption Condition	Board members need a baseline understanding of AI before meaningful and critical adoption is possible	I1, I2, I5, I6, I9, I10
	Risk of Superficial Adoption / Underutilization	Adoption can fail if the tool is not embedded into board routines: the board may use it superficially, or fail to exploit its full potential	I1, I4, I5, I6
	Resistance from Professional Identity and Habits	Experienced board members may resist new tools because professional identity is tied to having the answers	I2, I4, I5, I6, I9.

Appendix E: Full Design Principle Traceability Matrix

DP	Name	Specification	Empirical Grounding	Theoretical Grounding
DP1	Curation over comprehensiveness	Surface 3-5 strategic signals per briefing cycle, ranked by relevance and source credibility.	*Theme 1* (information volume causes paralysis, not clarity);	(Simon, 1955; Ward, 2023)
DP2	Plain language with layered depth	Executive summary by default; deeper detail accessible on demand.	*Theme 5* (layered format requirement stated independently by two interviewees; pedagogical clarity as explicit condition).	(Barredo Arrieta *et al.*, 2020; Rieger *et al.*, 2025)
DP3	Consistent pre-meeting baseline for all board members	The tool provides the same external intelligence report to all board members prior to each meeting.	*Theme 1* (unequal analytical backgrounds across board members due to individual supplementation behavior).	(Forbes and Milliken, 1999; Ben-Shahar *et al.*, 2025)
DP4	Strategic days as the primary integration point	Briefing is structured to align with the multi-stage strategic-day process, beginning with external scanning two weeks prior.	*Theme 3* (strategic days identified as the primary integration point in the governance year; external orientation of the strategic funnel as the entry point where management inputs are most constrained).	(Choo, 1994; Rohrbeck and Gemünden, 2011; Li *et al.*, 2018)
DP5	Contextual framing and instructional clarity	Outputs are thematically organized and pedagogically presented; every signal is framed with its strategic significance before delivery.	*Theme 5* (respondents uniformly stated raw data is insufficient; synthesis that produces meaning, not information requiring further synthesis, is the stated requirement).	(Daft and Weick, 1984; Choo, 1994)
DP6	Pre-meeting vs. during-meeting use	Primary use case is pre-meeting preparation; in-meeting use is acknowledged as a minority position	*Theme 5* (majority of interviewees stated pre-meeting as primary use case; time scarcity during meetings cited as the	(Simon, 1955; Forbes and Milliken, 1999).

		and treated as out of scope for the primary design.	determining constraint).	
DP7	Source and epistemic independence from the management channel	Tool scans external sources directly without reference to management's editorial perspective or management-prepared board documents.	*Theme 1* (pre-meeting material passes exclusively through the CEO, creating a structurally filtered epistemic horizon).	(Daft and Weick, 1984; McNulty and Pettigrew, 1999; Engstam *et al.*, 2024)
DP8	Weak signal detection and adjacent-industry scanning	Scanning extends beyond the immediate competitive landscape to adjacent sectors and emerging regulatory signals; at least one adjacent-industry indicator included per cycle regardless of the organization's current strategic frame.	*Theme 4* (adjacent-industry blind spot as a structural gap in current foresight practice); *Theme 3* (interviewees actively imported external perspectives to compensate for sector-level limitation during strategic days).	(Daft and Weick, 1984; Robinson and Simmons, 2018; Grove, Clouse and Xu, 2023)
DP9	Challenging perspectives and flagging issues	Tool surfaces alternative and opposing readings of strategic assumptions rather than confirming the management plan; presents a counter-analysis alongside each primary signal.	*Theme 4* (historical scenario work described as systematically optimistic and methodologically unable to surface what could go wrong); *Theme 5* (generative challenger use case tool stress-testing management strategy described independently by interviewees).	(Forbes and Milliken, 1999; Rohrbeck and Gemünden, 2011; Ben-Shahar *et al.*, 2025)
DP10	Continuous monitoring with configurable schedule	Continuous environmental monitoring delivered on a board-configured cadence; tool does not operate only on a pre-meeting trigger.	*Theme 2* (episodic governance cycle does not correspond with the pace of strategic disruption; strategy described as outdated from the moment of approval).	(Haarhaus and Liening, 2020; Engstam *et al.*, 2024)

DP11	Scenario building and consequence mapping	Generation of at least three scenarios (baseline, optimistic, pessimistic) with traced second-order consequences across two or more organizational processes, including early warning indicators.	*Theme 4* (scenario gap: historical scenarios were optimistic, infrequent, and provided only probability estimates without consequence mapping).	(Haarhaus and Liening, 2020; Marinković *et al.*, 2022)
DP12	Source linking for every claim	Each output entry includes a direct, clickable link to the originating source document; no assertion is generated without a traceable source.	*Theme 6* (source transparency identified as the single most consistently stated trust requirement across all ten participants).	(Barredo Arrieta *et al.*, 2020; Zeiser, 2024)
DP13	Confidence ratings	Each output entry carries a three-level confidence indicator (High / Medium / Low); Low-rated items include a plain-language explanation of the reason for low confidence.	*Theme 6* (confidence ratings endorsed by interviewees as a verification mechanism; multi-source threshold described as existing practice in one management group).	(Rieger *et al.*, 2025)
DP14	Parallel-use design	Tool introduced as an addition to existing board preparation activities for a minimum six-to-twelve-month trust accumulation period; exclusive use is not required.	*Theme 6* (trust described as a temporal achievement; interviewees described a transition cycle of verifying AI outputs alongside own research before reducing parallel preparation).	(Fahnenstich, Rieger and Roesler, 2024; Marocco, Barbieri and Talamo, 2024)
DP15	Public data by default	Tool operates on public sources by default; integration of internal data sources requires an explicit board resolution and a defined security protocol.	*Theme 8* (data security raised as an adoption barrier; one participant equated data strategy to corporate strategy, framing internal data access as a governance decision, not a	(Chriki *et al.*, 2025; Papagiannidis, Mikalef and Conboy, 2025)

			technical one); *Theme 6* (data privacy as a trust-architecture dimension).	
DP16	Structured deployment with change management protocol	Tool introduction is accompanied by structured onboarding, calendar integration, and a defined pilot phase; it is not deployed as a standalone IT rollout.	*Theme 8* (absence of behavioral change identified as the primary cause of deployment failure; tool adoption without process integration leads to abandonment).	(Davis, 1989; Venkatesh *et al.*, 2003)
DP17	Augmented intelligence stage only	Tool produces analytical input only; it issues no recommendations, evaluations, or strategic conclusions; all decision authority remains with the board.	*Theme 7* (board members uniformly distinguished between what AI can supply and what the board is accountable for; fiduciary responsibility described as non-delegable).	(Mertens, 2023; Ahdadou, Aajly and Tahrouch, 2024; Agnese, Arduino and Di Prisco, 2025; Ahdadou *et al.*, 2025)
DP18	Strategic framing constraint	Outputs are scoped to board-level strategy concerns; the chair defines the boundary between board-relevant strategic content and management-relevant operational content during implementation.	*Theme 3* (external strategic funneling identified as the board's domain); *Theme 7* (interviewees consistently stated that the tool must not overstep into operational matters).	(McNulty and Pettigrew, 1999)
DP19	Named internal accountability	A single named individual in an information or digital officer capacity is accountable for tool operation, outputs, and maintenance; the CEO retains accountability for reviewing content before board delivery.	*Theme 8* (named accountability owner identified as a prerequisite for trust; absence of a named owner described as a sufficient condition for non-use).	(Chriki *et al.*, 2025)

DP20	Board AI literacy as governance infrastructure	Baseline AI competence is required across all directors; addressed through a structured onboarding process for current members and inclusion of AI competence in future board nomination criteria.	*Theme 8* (AI literacy identified as a key precondition; lack of literacy produces reflexive rather than critical challenge of AI outputs).	(Marocco, Barbieri and Talamo, 2024)
DP21	Governance documentation and audit trail	Tool maintains a full audit log covering data sources accessed, outputs generated, delivery timestamps, CEO curation completion, and references to outputs in board decision records.	*Theme 7* (directors emphasized individual accountability for any AI-informed governance decision; ability to demonstrate that AI analysis was reviewed critically, not accepted passively, is a legal requirement under the Business Judgment Rule).	(Chriki *et al.*, 2025)

TRITA – ITM-EX 2026:215
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